

Bayesian-Optimization-Guided Discovery of NiTi(Co,Cu,Pd,Hf,Zr) Multi-Principal Element High-Temperature Shape Memory Alloys

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Abstract

Designing high-temperature shape memory alloys (HTSMAs) requires balancing competing objectives — transformation temperature, thermal hysteresis, latent heat, and recoverable transformation strain — and the multi-principal-element (MPE) alloy space in which the most promising HTSMAs now lie is too large for exhaustive exploration. This work reports a closed-loop, multi-objective batch Bayesian optimization campaign in the NiTi(Co,Cu,Pd,Hf,Zr) system, spanning quaternary to septenary compositions. Across three iterations of experiment-informed acquisition, 87 alloys were synthesized and characterized, and the cumulative normalized hypervolume of the target property space rose from 0.544 to 0.808, corresponding to a 48.5% expansion of the accessible trade-off region. The best-performing composition, Ni₄₆Ti₂₈Co₂Pd₂Hf₂₂, achieved a maximum transformation strain of 4.18% at 300–400 MPa together with a thermal hysteresis of 16.5 °C in stress-free DSC screening and 13.9 °C under 50 MPa — a combination that places it above the empirical strain–hysteresis Pareto front in the >200 °C martensite-start regime. These results show that closed-loop Bayesian optimization, coupled with high-throughput characterization and targeted mechanical validation, can navigate complex HTSMA design spaces and identify compositions that jointly achieve high transformation temperatures, low hysteresis, and large recoverable strains.

Keywords: High-Temperature Shape Memory Alloys, Martensitic Phase Transformation, Transformation Strain, Enthalpy of Transformation, Thermal Hysteresis, Bayesian Optimization

1. Introduction

Shape memory alloys (SMAs) are functional materials whose actuation and energy-dissipation capabilities originate from a reversible, diffusionless martensitic transformation between high-symmetry austenite and low-symmetry martensite phases [1–3]. Because the transformation couples strongly to temperature and stress, SMAs enable compact actuators and adaptive structures with sustained interest across biomedical devices, aerospace structures, and industrial components [4–8]. Each application requires a distinct set of properties, including thermal properties of transforming phases, transformation temperatures and thermal hysteresis, ΔT , and mechanical behavior, transformation strain (ε_{tr}) and irrecoverable strain (ε_{irr}) [3, 9–11]. HTSMA behavior is governed by both composition and microstructure, and modifications to either can simultaneously affect multiple functional and

mechanical properties, often leading to competing performance trade-offs.

Near-equiatomic NiTi alloys are the most widely used SMA because they offer an attractive combination of strength, ductility, corrosion resistance, and recoverable strain. However, their transformation temperatures are limited to 100 °C, restricting applications requiring phase transformation at elevated temperatures [12]. High-temperature SMAs (HTSMAs) rely on alloying strategies to increase transformation temperatures but often show increased functional degradation, reduced cyclic stability, and lower ductility [1, 13–16]. Consequently, HTSMA development is a multi-objective design problem in which transformation temperatures must be increased while maintaining favorable thermal and mechanical performance.

In earlier studies, precious-metal additions (i.e., Pd and Pt) have been the most widely used strategy for increasing transformation temperatures in NiTi-based HTSMAs. Ternary NiTiPd and NiTiPt alloys can achieve transformation temperatures exceeding $\sim 700^\circ\text{C}$ [17–19]. However, such temperatures require substantial alloying additions ($> 15\text{ at.}\%$), significantly increasing material cost. The

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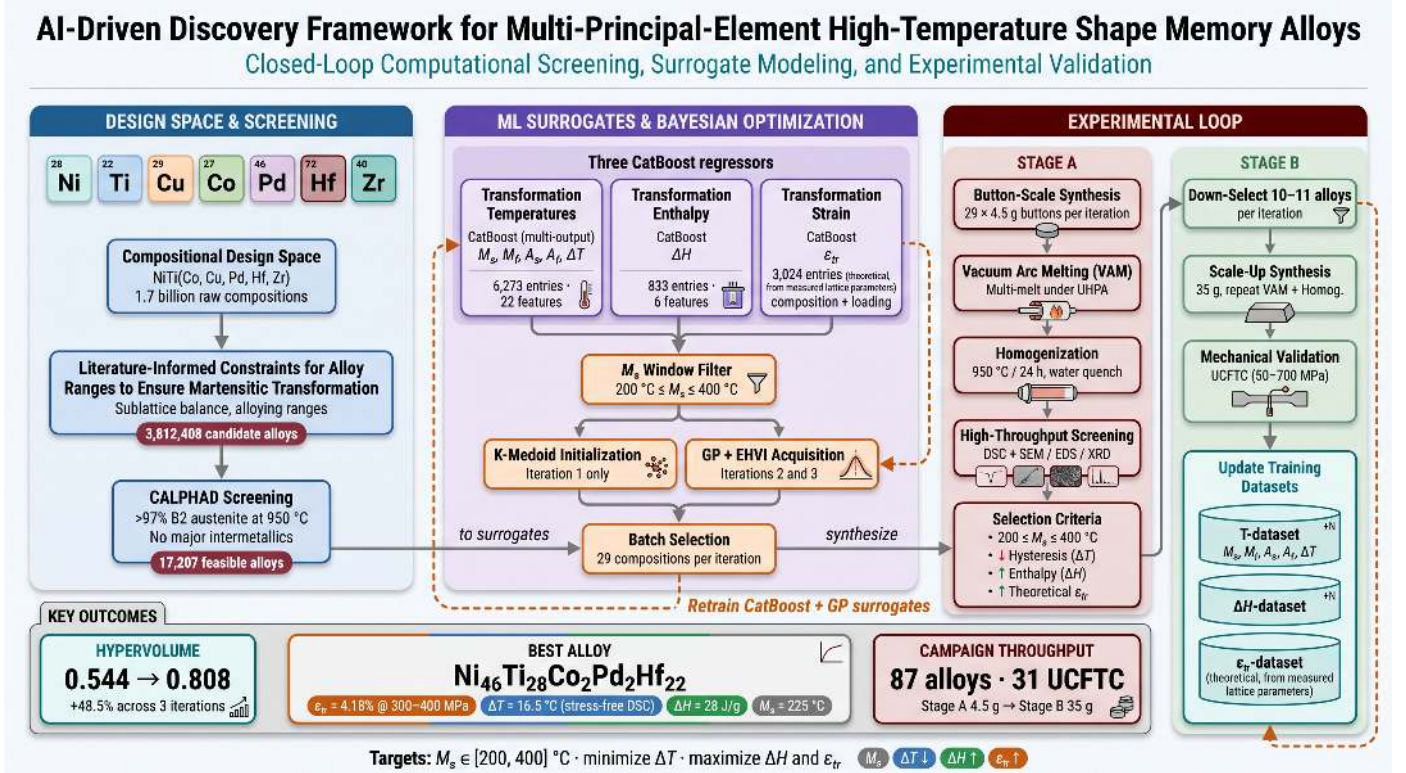


Figure 1: Integrated AI-driven framework for multi-principal-element high-temperature shape memory alloy discovery. The workflow combines literature-informed design-space definition, CALPHAD phase-stability screening, machine-learning (ML) property prediction, Bayesian optimization, and closed-loop experimental validation. Starting from the NiTi(Co, Cu, Pd, Hf, Zr) compositional design space, candidate alloys are down-selected using phase stability, ML-predicted transformation-temperature constraints, and multi-objective optimization targeting high transformation enthalpy (ΔH) and transformation strain (ϵ_{tr}) with low thermal hysteresis (ΔT). Selected compositions are synthesized as 4.5 g buttons, characterized thermally and structurally, down-selected for 35 g scale-up, and mechanically evaluated by UCFTC. The new experimental data are returned to the SMA database to retrain models and guide subsequent optimization iterations. Figure generated using Claude and Gemini and subsequently reviewed and edited by the authors.

compositional dependence is further complicated by a non-monotonic response, with transformation temperatures initially decreasing at low Pd/Pt contents (up to ~ 5 at.%) before rising at higher concentrations [12, 17, 18, 20, 21]. Despite these complexities, Pd and Pt offer important benefits, including reduced ΔT and improved cyclic stability [18, 20, 22–26]. Nevertheless, elemental cost remains a practical barrier limiting potential applications.

Hf and Zr have emerged as alternatives to precious-metal additions for elevating transformation temperatures in NiTi-based HTSMAs [27–43]. Like Pd and Pt, both elements initially suppress transformation temperatures at low concentrations below 10 at.%, followed by a near-linear increase at higher additions, enabling martensite start temperatures (M_s) exceeding 500 °C [12, 44, 45]. In these systems, transformation behavior is strongly coupled to thermal and processing history because of the rapid formation of coherent nano-precipitates. In Ni-rich NiTiHf compositions, nanoscale H-phase precipitates form rapidly during aging and alter transformation behavior through both local compositional changes and interactions with martensite growth. These precipitates can also improve dimensional stability by suppressing dislocation plasticity and reducing functional degradation during cycling

[27, 42, 43, 46–48]. This level of microstructural control provides an additional pathway for tailoring HTSMA performance beyond composition alone.

NiTiZr alloys show similar precipitation behavior, with nanoscale H-phase formation influencing transformation temperatures, ΔT , and recoverable ϵ_{tr} [49, 50]. However, differences in precipitation kinetics between Zr- and Hf-containing alloys shift precipitate growth and coarsening behavior, leading to distinct microstructure-property relationships. As a result, alloy design in the NiTi(Hf,Zr) system requires simultaneous control of chemistry, thermal history, and thermomechanical processing. For comparable alloy contents and processing conditions, Hf-containing alloys generally show higher transformation temperatures than their Zr-containing counterparts, whereas Zr-containing alloys often provide comparable actuation performance at lower cost and density and, in some quaternary alloys, improved dimensional stability [51].

Not all alloying additions are intended to increase transformation temperatures. Elements such as Cu are introduced to reduce thermal hysteresis, modify the transformation path and martensite phase structure, and improve functional stability. At sufficiently high concentrations, Cu stabilizes orthorhombic $B19$ martensite rather than

the monoclinic $B19'$ martensite characteristic of NiTi and NiTi(Hf,Zr) alloys, leading to improved lattice compatibility and cyclic stability, although typically at or below room temperature [15, 52–57]. Other additions, including Co, have been explored to tailor phase stability, transformation behavior, and thermomechanical response through changes in transformation temperatures and phase constitution [58–62]. Therefore, these observations highlight that HTSMA design requires balancing transformation temperatures with functional and mechanical performance.

Multi-principal-element (MPE) SMAs have emerged as a promising strategy for expanding the design space of HTSMAs beyond traditional ternary and quaternary compositions. Inspired by broader developments in high-entropy and compositionally complex alloys [63, 64], MPE HTSMAs seek to exploit increased chemical complexity to achieve combinations of transformation temperatures, hysteresis, and thermo-mechanical stability that may not be attainable in conventional SMA systems. Early studies by Firstov and coworkers [65–67] established the thermodynamic rationale for MPE SMA design and demonstrated the feasibility of ZrCuNiCo- and TiZrHfCoNiCu-based systems as SMAs. Subsequent investigations have expanded the compositional landscape and explored the effects of thermomechanical processing, transformation behavior, and manufacturability across several MPE alloy families [68–76].

Despite these advances, the functional performance of many reported MPE HTSMAs remains inferior to that of precipitation-strengthened NiTiHf alloys. Although elevated transformation temperatures have been demonstrated, many MPE compositions show relatively low recoverable transformation strains (typically $< 2\%$) and large thermal hysteresis, limiting their competitiveness for solid-state actuation applications [69, 71, 74–77]. Most studies have focused primarily on compositional effects, while the roles of precipitation, microstructural evolution, and thermomechanical processing have received comparatively less attention. As demonstrated in conventional NiTiHf and NiTiZr systems, these factors strongly influence transformation temperatures, hysteresis, cyclic stability, and transformation strain, suggesting that composition alone is unlikely to fully exploit the potential of MPE HTSMAs. Nevertheless, establishing composition–property relationships under controlled processing conditions represents a necessary first step toward broader composition–processing optimization.

The main challenge in MPE HTSMA development is the rapid expansion of the compositional design space with increasing chemical complexity. For example, expanding a ternary NiTiHf system to a quinary NiTiHfZrPdCu system increases the number of potential compositions by several orders of magnitude. This combinatorial expansion is compounded by the extreme composition sensitivity of these alloys: in NiTi-based systems a change of less than ~ 0.5 at.% in the Ni/Ti ratio can shift the martensite start

temperature from well below room temperature (or from no detectable transformation at all) to $\sim 200^\circ\text{C}$ [12, 78]. Combined with stringent phase-stability requirements, the prior probability that an arbitrary composition is simultaneously phase-pure, transformation-capable, and within the target M_s window is therefore extremely low: in the NiTi(Co,Cu,Pd,Hf,Zr) system explored here only $\sim 0.45\%$ of candidate compositions pass CALPHAD phase-purity screening ($> 97\%$ single-phase B2 austenite at the homogenization temperature), and within that pre-filtered subset transformability under a fixed thermal protocol is not guaranteed. Random or uninformed sampling of MPE chemistry is therefore prohibitively inefficient, and traditional trial-and-error exploration becomes impractical. This stands in contrast to many high-entropy alloy (HEA) optimization problems, where the disordered solid-solution region of interest typically spans a substantially larger and more contiguous fraction of the compositional space; the MPE HTSMA search problem is therefore fundamentally harder than HEA optimization at a comparable nominal dimensionality. Consequently, computational approaches such as machine learning (ML), Bayesian optimization (BO), and CALPHAD-based phase-stability screening have emerged as promising tools for navigating these high-dimensional design spaces while accounting for competing performance objectives.

Recent demonstrations of ML-guided SMA discovery at near-room-temperature conditions have highlighted the potential of data-driven alloy design [79]. However, similar frameworks remain comparatively unexplored for HTSMAs [80–82], where multi-objective optimization is complicated by competing relationships among transformation temperatures, hysteresis, transformation strain, and functional stability. Although several studies have reported compositionally complex alloys with elevated austenite finish (A_f) or M_s temperatures, improvements in temperature capability are often accompanied by increases in hysteresis, reductions in recoverable strain, or degradation in mechanical performance [80, 81, 83]. Furthermore, experimental validation has been limited to a small fraction of the accessible design space, making it difficult to assess whether reported compositions approach true multi-objective optima or represent locally favorable solutions. These limitations motivate the development of integrated computational–experimental frameworks capable of simultaneously navigating large compositional spaces, accounting for competing performance objectives, and experimentally validating predicted alloy candidates.

In this study, a multi-objective batch Bayesian optimization (BBO) framework, as depicted in Fig. 1, identified novel MPE HTSMA compositions targeting $M_s > 200^\circ\text{C}$ while simultaneously maximizing transformation enthalpy (ΔH) and ϵ_{tr} and minimizing ΔT . The approach combines CALPHAD-based phase-stability screening, ML-assisted surrogate modeling, and iterative Bayesian optimization with high-fidelity experimental validation of selected candidates. Transformation temperatures, ΔH , and ΔT were

measured through high-throughput thermal cycling, while ε_{tr} was assessed using (i) a high-throughput predictive approach based on austenite and martensite lattice parameters, and (ii) selected experimental validation. By coupling computational down-selection with thermal and mechanical evaluation of shape-memory behavior, this work addresses a key limitation of existing MPE HTSMA design studies, which often emphasize unloaded thermal behavior without demonstrating the corresponding mechanical functionality for the proposed compositions [80, 81, 83].

2. Methods

2.1. Overview and Objectives

This work explored a high-dimensional compositional space within the NiTi(Co, Cu, Pd, Hf, Zr) alloy family, spanning quaternary through septenary chemistries and comprising 3,812,408 unique candidate compositions. The optimization targets are to maximize ΔH and ε_{tr} while minimizing ΔT . Transformation enthalpy can be used as a DSC-measurable proxy for both the completeness of the martensitic transformation and the fraction of transformation-capable material: low ΔH indicates either a weakened B2 $\leftrightarrow$ B19/B19' transformation or a large fraction of untransformable (non-B2) secondary phase. Either failure mode gives small recoverable strain, degraded cyclic stability, and unreliable actuation, making ΔH an appropriate early-rejection criterion. To target high-temperature actuation, candidates are furthermore required to satisfy a functional operating constraint with $M_s \in [200, 400]^\circ\text{C}$.

To evaluate these competing objectives at scale, a four-stage workflow was used:

1. Explicit design-space construction informed by established alloying trends.
2. CALPHAD-based thermodynamic screening to remove compositions unlikely to be processable or stable during homogenization.
3. Property prediction via physics-informed machine-learning (ML) models developed in prior work [84–86] and adapted here to the expanded chemical domain.
4. Surrogate-assisted multi-objective Bayesian optimization (BO) to prioritize experimental evaluation.

The high-throughput thermal characterization enabled evaluation of relatively large experimental batches. In contrast, the lower throughput and higher cost associated with thermo-mechanical testing, especially the uniaxial constant-force thermal cycling to evaluate transformation strains, required additional down-selection. Accordingly, each batch was reduced from 29 alloys to 10 candidates in the first and third iterations and 11 candidates in the second iteration using experimentally measured thermal properties together with crystallographically informed estimates of transformation strain. The

iterative cycle of alloy selection, synthesis, characterization, and model retraining continuously expanded the experimental database, improved predictive accuracy, and guided the BBO framework toward regions of increasing Pareto-optimal hypervolume.

2.2. Experimental Methods

Alloy buttons (4.5 g) were synthesized by vacuum arc melting using high-purity feedstock: Ni (99.98%), Ti (99.98%), Cu (99.995%), Co (99.95%), Pd (99.999%), Hf (99.9% with 2% nominal Zr), and Zr (99.5% with 3% nominal Hf). Melting was conducted under ultra-high-purity argon (UHPA, 99.999%). The chamber was evacuated to 5×10^{-5} Torr and backfilled with UHPA. The purge cycle was repeated at least five times prior to first melting. Each button was melted, flipped, and re-melted a minimum of five additional times to improve chemical homogeneity. Elemental accuracy was quantified by the difference between the initial elemental charge weight and the final synthesized alloy weight. Alloys with $< 0.1\%$ mass loss were considered sufficiently compositionally accurate for characterization, whereas alloys exceeding this threshold were re-melted from new elemental feedstock until the criterion was met.

Buttons were encapsulated in quartz tubes under argon. Tubes were evacuated to 3×10^{-2} Torr and refilled with UHPA. This process was repeated a minimum of five times. High vacuum was achieved using a turbomolecular pumping system to reach 5×10^{-5} Torr, after which the tubes were backfilled to 13 Torr with UHPA and sealed. Solution heat treatments were conducted at 950°C for 24 h, followed by rapid water quenching. Quenching was performed by removing the tubes from the furnace, immediately immersing them in water, and cracking the tubes to promote rapid cooling.

Specimens for thermal analysis were prepared by electrical discharge machining (EDM): 1 mm thick slices were sectioned from each button and 3 mm disks were extracted from these slices. Differential scanning calorimetry (DSC) was performed on a TA Instruments DSC 2500 at a constant heating/cooling rate of $10^\circ\text{C min}^{-1}$. Per-alloy compositions, processing conditions, DSC-derived transformation temperatures and enthalpies, lattice parameters, and ML-predicted properties for all alloys generated in the campaign are tabulated in the accompanying spreadsheet (Supplementary Information S1). Each alloy was thermally cycled twice, and only the second cycle was analyzed to avoid first-cycle transient effects, with transformation temperatures (M_s , M_f , A_s , A_f) extracted by the tangent-line intercept method per ASTM F2004. The DSC accommodates up to 50 samples and completes their measurements automatically without further operator intervention.

Microstructural characterization was performed using a Thermo Fisher Scientific Quanta 600 FEG scanning electron microscope (SEM) equipped with an Oxford Instruments energy-dispersive X-ray spectroscopy (EDS) detec-

tor. Secondary-phase area fractions in the micrographs shown in this work were measured in ImageJ [87]. Room-temperature and *in situ* high-temperature X-ray diffraction (HT-XRD) were performed using a Bruker D8 Discover diffractometer equipped with a Bruker VANTEC-500 two-dimensional (2D) area detector. Measurements were conducted using Cu $K\alpha$ radiation ($\lambda = 1.54 \text{ \AA}$) over a 2θ range of 20° to 80° at a scan rate of 2° min^{-1} . For high-temperature structural characterization, an MRI Physikalische Geräte GmbH environmental heating stage equipped with a platinum heating strip was used. To minimize sample oxidation at elevated temperatures, the stage was operated under vacuum and enclosed by an X-ray-transparent beryllium (Be) dome. Bulk specimens were loaded in multi-specimen batches, enabling sequential diffractogram collection without breaking the environmental seal or thermally cycling the chamber between individual measurements. Initial target temperatures for structural measurements were established using prior differential scanning calorimetry (DSC) data. To isolate the high-temperature B2 phase, the stage was heated above the anticipated austenite finish (A_f) temperature. If residual martensitic reflections were observed, the temperature was incrementally increased by $10\text{--}15^\circ\text{C}$ until a fully austenitic structure was verified. Subsequently, the samples were cooled to a target temperature below the anticipated martensite finish (M_f). If residual austenite peaks persisted, the temperature was further reduced in $10\text{--}15^\circ\text{C}$ decrements until a fully martensitic (B19') structure was isolated. Following data collection, the two-dimensional diffraction frames were integrated into standard one-dimensional intensity versus 2θ patterns. Final phase identification and lattice parameter extraction were performed from the integrated diffraction profiles using Le Bail whole-pattern decomposition [88].

Selected compositions were produced at larger scale (35 g) using the same melting and heat-treatment protocol and the same $< 0.1\%$ mass-loss criterion. A 1 mm slice was removed from the end of each ingot by EDM to extract 3 mm disks for DSC verification. If the DSC-derived transformation temperatures and ΔT for the scaled alloy overlapped those of the corresponding button within $\pm 50^\circ\text{C}$, the composition proceeded to mechanical testing.

Thermo-mechanical characterization was conducted through uniaxial constant-force thermal cycling (UCFTC) on a servo-hydraulic MTS 810 tensile test machine. Specimen heating was achieved conductively using 350 W wraparound heating collars with a maximum output of 650°C , attached around metal jackets press-fit to the grips and surrounded by fiberglass insulation. The same arrangement was used for both the top and bottom grips to minimize the temperature gradient through the specimen. The jackets were custom-built from 3600 brass to serve as both heating and cooling source. Custom-bored holes channel liquid nitrogen through the jackets to cool the specimen during the cooling segments of the thermal cycle. Grip temperatures were measured using K-type ther-

mocouples held in contact by secondary mounting pieces near the specimen/grip contact points. Specimen temperature was measured with a K-type thermocouple attached to the specimen near the midpoint of the gauge section. To remain consistent with DSC cycling performed under no load, a sample heating and cooling rate of $10^\circ\text{C min}^{-1}$ was used. An MTS high-temperature extensometer directly attached to the gauge section of the specimens was used to measure the uniaxial strain.

Scaled-up alloys were machined by EDM into tensile specimens with a 26 mm overall profile, a thickness of 1 mm, and a gauge section measuring $8 \text{ mm} \times 3 \text{ mm} \times 1 \text{ mm}$. ε_{tr} under load was measured by uniaxial constant-force thermal cycling (UCFTC). Specimens were initially heated from room temperature to the upper cycle temperature (UCT) under 10 MPa, where

$$\text{UCT} = \lceil A_f + 150^\circ\text{C} \rceil_{50}, \quad (1)$$

and $\lceil \cdot \rceil_{50}$ denotes rounding up to the nearest 50°C . The load was then increased to 50 MPa and the sample was thermally cycled between UCT and a lower cycle temperature (LCT),

$$\text{LCT} = \lfloor M_f - 80^\circ\text{C} \rfloor_{50}, \quad (2)$$

where $\lfloor \cdot \rfloor_{50}$ denotes rounding down to the nearest 50°C . This cooling/heating cycle was repeated at successive loads of 50, 100, 150, 200, 300, 400, and 500 MPa (or until fracture). For specimens that survived 500 MPa, the load was reduced to 10 MPa, the sample was cooled to room temperature, and then heated to $(\text{UCT} + 50^\circ\text{C})$. Additional cycling at 600 MPa and 700 MPa was performed (until failure) using the temperature range LCT to $(\text{UCT} + 50^\circ\text{C})$.

Loaded transformation temperatures and strain metrics were extracted from UCFTC strain-temperature curves using the tangent-intercept procedure illustrated in Fig. 2. Briefly, linear baselines were fit to the pre-/post-transformation segments to represent thermal expansion/contraction of the austenite and martensite phases. Tangent lines were then fit to the approximately linear portions of the transformation segments during cooling and heating; intersections of these tangents with the corresponding phase baselines define (M_s, ε_{M_s}) , (M_f, ε_{M_f}) , (A_s, ε_{A_s}) , and (A_f, ε_{A_f}) . In addition, temperatures/strains marked with “*” denote onset/finish based on the first deviation from (and final return to) the relevant baseline.

Transformation strain was defined as

$$\varepsilon_{\text{tr}} = \varepsilon_{A_s} - \varepsilon_{\text{UCT}}, \quad (3)$$

and residual strain as

$$\varepsilon_{\text{res}} = \varepsilon_{\text{UCT}} - \varepsilon_i, \quad (4)$$

where ε_i is the strain at the beginning of the loaded cycle in

the fully austenitic state, and ϵ_{UCT} is the strain measured at UCT. ΔT was computed using midpoint temperatures,

$$M_{50} = \frac{M_s + M_f}{2}, \quad A_{50} = \frac{A_s + A_f}{2}, \quad \Delta T = A_{50} - M_{50}. \quad (5)$$

Definitions follow ASTM E3097-23 [83]. Raw UCFTC strain–temperature curves and tangent-intercept analyses for the 31 scaled-up alloys characterized in this campaign are compiled in Supplementary Information S2.

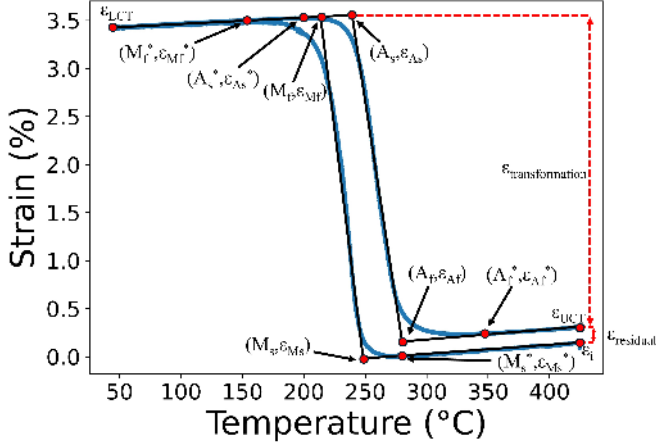


Figure 2: Schematic illustration of the tangent-intercept methodology used to determine transformation temperatures under constant force and strain metrics from uniaxial constant-force thermal cycling (UCFTC) strain–temperature responses. Linear baselines representing the thermal expansion and contraction behavior of the austenite and martensite phases are intersected with tangents fitted to the transformation regions during heating and cooling to identify the characteristic transformation temperatures (M_s , M_f , A_s , and A_f) and their corresponding strain values. The procedure is also used to quantify transformation strain (ϵ_{tr}), residual strain (ϵ_{res}), and thermal hysteresis (ΔT) [83].

2.3. Design Space Definition

Because the effects of alloying in NiTi-based SMAs are highly non-linear and strongly composition-dependent, the compositional design space was constructed using trends established in lower-complexity binary, ternary, and quaternary systems. The objective was to balance broad exploration of the HTSMA design space with physically informed constraints that preserve martensitic transformation and avoid compositions associated with undesired phase formation (such as those that would lead to excessive formation of Ti_2Ni -type phases) or degraded functional response. Starting from the NiTi system with candidate alloying additions of Co, Cu, Pd, Hf, and Zr, the unconstrained compositional space contained approximately 1.7 billion possible compositions. Application of literature-informed compositional constraints and elemental down-selection reduced this space to 3,812,408 candidate alloys suitable for further evaluation, as detailed below. The full candidate list can be regenerated with a script in the public repository [89].

All alloying additions were parameterized as substitutions on either the Ni or Ti sublattice. To promote

retention of a transformable B2 austenite matrix, near-stoichiometric balance was enforced on the grouped sublattices. The Ti-site group (Ti + Hf + Zr) was constrained to 48.5–50 at.% to minimize Ti_2Ni -type precipitates, while the Ni-site group (Ni + Cu + Co + Pd) was constrained to 48.5–51.5 at.% to avoid full suppression of martensitic transformation and the appearance of strain-glass transitions. These bounds favor transformation to B19/B19'-type martensite while maintaining sufficient flexibility for compositional optimization. All candidate alloys additionally satisfied the global atomic balance constraint:

$$\text{Ni} + \text{Ti} + \text{Cu} + \text{Co} + \text{Pd} + \text{Hf} + \text{Zr} = 100 \text{ at.}\%. \quad (6)$$

Cu was considered a substitutional element on the Ni sublattice and sampled at either 0 at.% or 5–25 at.% in 1 at.% increments. When present, a minimum concentration of 5 at.% was imposed to ensure a measurable influence on transformation behavior and ΔT [54, 57, 79, 90, 91]. At higher concentrations, commonly reported above approximately 15 at.%, Cu additions promote a transition from monoclinic B19' martensite to orthorhombic B19 martensite, which has been associated with improved crystallographic compatibility and reduced hysteresis [15, 53, 54]. Because the stability of these transformation paths remains sensitive to operating temperature, Cu was incorporated as a design variable for hysteresis reduction without assuming any positive or negative effect across the targeted M_s range.

Co was treated as an optional Ni-site substitution and sampled at 0–5 at.% in 1 at.% increments. This restricted range reflects its influence on both transformation thermodynamics and mechanical response. Previous studies have reported approximately linear reductions in transformation temperatures of NiTi-based SMAs within this concentration range together with improved resistance to plastic deformation [60, 61, 92]. Limiting Co to low concentrations therefore preserves consistency with reported behavior while reducing the likelihood of generating compositions incompatible with the targeted high-temperature operating window and causing formation of unexpected second-phase particles.

Pd was sampled as a Ni-site substitution over a continuous range of 0–25 at.% in 1 at.% increments. Pd shows a non-monotonic influence on transformation temperatures: low concentrations tend to suppress transformation temperatures, whereas higher concentrations produce substantial increases while maintaining relatively narrow ΔT [19, 93]. Because the present optimization simultaneously targets elevated transformation temperatures and reduced ΔT , Pd was retained across the full composition range.

Hf and Zr were treated as Ti-site substitutional elements and sampled at either 0 at.% or 5–25 at.% in 1 at.% increments. Both elements are used to increase transformation temperatures in NiTi-based HTSMAs. However, at low concentrations (< 5–10 at.%), both systems may

show an initial suppression of transformation temperatures, depending on the Ni richness of the alloy, before the temperature-increasing effect becomes dominant at higher alloy contents [12, 44, 94]. Accordingly, a minimum concentration of 5 at.% was imposed whenever Hf or Zr was present to focus exploration on compositions more likely to show high transformation temperatures.

Ni and Ti concentrations were sampled more finely to capture the strong sensitivity of transformation behavior to deviations from stoichiometry. Ni was allowed to vary between 20 and 51.5 at.% and Ti between 20 and 50 at.%, each using 0.2 at.% increments. The upper Ni limit reflects the well-established suppression of transformation temperatures and transformation enthalpy in increasingly Ni-rich NiTi alloys. A finer discretization for Ni and Ti was therefore used to accurately resolve composition–property relationships near stoichiometry, whereas the remaining alloying additions were discretized in 1 at.% increments.

Finally, elemental complexity was defined solely by the presence of an element at a non-zero concentration within the discretized design space, without imposing an additional threshold concentration for inclusion. This definition enables consistent comparison among compositions spanning different levels of chemical complexity. The resulting workflow used to construct the compositional design space is summarized in Fig. 1.

2.4. CALPHAD-Based Thermodynamic Screening

The design space was further reduced using thermodynamic screening to remove compositions unlikely to be processable under the homogenization schedule. Equilibrium calculations were performed using the TC-Python API within Thermo-Calc (version 2023a) [95] with the TCNI12 database [96]. Each candidate composition was evaluated at the homogenization temperature of 950 °C, corresponding to the solution-treatment / homogenization schedule used throughout this study. Candidate compositions predicted to initiate melting at or below 950 °C were excluded to ensure stability during the 950 °C/24 h solution heat treatment. In addition, candidates predicted to contain significant fractions of intermetallic phases or to lack a majority (> 97%) single-phase B2 austenite at ≥ 950 °C were removed.

Application of these screening criteria reduced the candidate pool from 3,812,408 compositions to 17,207 feasible alloys, which served as the input space for subsequent machine-learning surrogates for transformation temperatures, thermal hysteresis, transformation enthalpy, and transformation strain. This reduction corresponds to a CALPHAD pass rate of $\sim 0.45\%$ (roughly 1 in 220 candidate compositions), underscoring the intrinsic rarity of phase-pure single-B2 austenite within the unconstrained MPE design space.

2.5. Machine Learning Surrogate Models

Property evaluation across the CALPHAD-filtered candidate space relied on machine-learning (ML) surrogate

models that map alloy chemistry and processing conditions to target properties.

The surrogate-model framework used in this campaign builds on three single-property ML predictors developed for NiTi-based SMAs in previous work by the authors: a CatBoost regressor for transformation temperatures ($R^2 = 0.95$, MAE = 17.4 °C, RMSE = 25.1 °C on a hold-out test set) [84]; a piecewise-linear phase-compatibility model for the middle eigenvalue λ_2 of the lattice deformation tensor ($R^2 = 0.91$, MAE = 0.01) [85]; and a Lattice Deformation Theory (LDT) CatBoost model for theoretical transformation strain ($R^2 = 0.96$, MAE = 0.60%, RMSE = 0.90%) [86]. Code and data for all three predictors are publicly available (<https://github.com/sinazadeh/CatBoost-SMAs>, <https://github.com/sinazadeh/Phase-Compatibility-Model-NiTi>, <https://github.com/sinazadeh/Transformation-Strain-Model-NiTi>).

These models are applied here largely as developed. The λ_2 and LDT models act as fixed deterministic filters during candidate down-selection, screening the CALPHAD-feasible set toward compositions with favorable B19/B19' pathways ($\lambda_2 \rightarrow 1$) and adequate theoretical ε_{tr} ($\geq 2.5\%$); the transformation-temperature and ΔH CatBoost predictors enter the closed loop as the $M_s/M_f/A_s/A_f/\Delta T$ and ΔH sources for the BO ranking and are incrementally retrained on the appended batch data after each iteration (Section 2.5.1). No training corpus at comparable scale yet exists for the full MPE chemistry, so the NiTi-trained models are used as priors rather than fit to MPE-specific data from scratch. Individual surrogate accuracy is expected to degrade in absolute terms when extrapolating from binary/ternary NiTi training data to the higher-dimensional MPE composition space. This degradation does not strictly disqualify the surrogates from use in BO, however: the goal of BO is to identify the composition that optimizes the objective(s) rather than to predict the objective value at every point in the design space, and what matters for that goal is preservation of rank-ordering among promising candidates rather than pointwise accuracy. Whether the integrated CALPHAD + ML + BO stack achieves this on the MPE chemistry can be assessed at the campaign level: if the framework is operating as intended, the closed-loop search should produce progressively higher yields against the screening targets and progressively wider coverage of the Pareto front from one iteration to the next.

Inputs to the ML property models included (i) elemental compositions, (ii) composition-derived chemical and physical descriptors, and (iii) processing parameters relevant to the property being modeled. For the transformation strain ε_{tr} model, the input space was further augmented with the mechanical loading condition (tension or compression) to capture the tension/compression asymmetry of martensitic transformation. The overall modeling workflow comprised dataset curation, descriptor construction, feature selection, and model training, with model architectures selected to enable accurate inference on heteroge-

neous tabular data.

Because microstructure and functional response are strongly influenced by processing history, all candidate alloys were synthesized and evaluated using a common processing schedule to isolate compositional effects. Simultaneous optimization of composition and processing would substantially increase both search dimensionality and experimental burden, and was therefore beyond the scope of the present study. Accordingly, homogenization temperature and duration were fixed throughout the Bayesian optimization (BO) campaign, and all processing-related model inputs were held constant during candidate evaluation. Predicted properties and Pareto comparisons should therefore be interpreted as composition-dependent responses evaluated under a common processing condition.

2.5.1. Data Preparation and Curation

Three property-specific datasets were assembled from the literature as well as accumulated in-house measurements from previous experimental campaigns [82, 84–86] to train the initial ML predictors. All records were standardized through unit harmonization, consistent labeling, and removal of incomplete or non-physical entries. The resulting baseline datasets were:

- **Transformation enthalpy:** 833 experimentally measured entries for NiTi-based SMAs.
- **Transformation strain:** 3,024 theoretical strain responses derived from 126 distinct compositions under tensile and compressive loading.
- **Transformation temperatures and thermal hysteresis:** 6,273 experimental entries containing M_s , M_f , A_s , A_f , and ΔT .

During the closed-loop campaign, the transformation-temperature measurements (M_s , M_f , A_s , A_f , ΔT) and transformation enthalpies collected in each experimental batch were appended to the temperature and enthalpy datasets, respectively, and the transformation-temperature and ΔH CatBoost predictors were retrained prior to the next iteration. The λ_2 phase-compatibility model and the LDT-based theoretical transformation-strain surrogate produce composition-derived deterministic outputs and are applied throughout the campaign as fixed filters without retraining (Section 2.5).

2.5.2. Descriptor Construction and Feature Selection

Each composition was encoded using composition-derived chemical and physical descriptors generated with the Composition-Based Feature Vector (CBFV) package [97–102]. The featurization process generated 5,770 statistical descriptors spanning multiple descriptor families, including Jarvis, Magpie, Oliynyk, Mat2vec, OneHot, and HEACalculator.

Feature selection proceeded in two stages to reduce redundancy and improve generalization. First, Pearson correlation filtering removed one feature from each highly correlated pair to mitigate multicollinearity. Second, SHapley Additive exPlanations (SHAP)-based recursive feature elimination pruned features with persistently low importance based on Shapley-value attributions, yielding compact models that emphasize physically meaningful descriptors while retaining predictive accuracy.

2.5.3. Model Architecture and Training

CatBoost regressors were selected as the primary predictive models because of their strong performance on heterogeneous tabular datasets, reducing sensitivity to extensive tuning in early model-development stages. This choice is consistent with benchmarking studies of gradient-boosting frameworks, which report CatBoost and XGBoost among the strongest baseline performers for materials-property prediction tasks [103], while all major implementations perform competitively after hyperparameter optimization. In addition, CatBoost provides practical support for multi-target prediction (either via multi-target objectives or by training one model per target) and includes built-in tools for feature importance and feature selection (including SHAP-based attributions), which were used in the correlation-filtering and SHAP-guided elimination workflow. Models were further refined via hyperparameter optimization to minimize validation loss, with L2 regularization and early stopping used to limit overfitting and improve robustness when evaluating CALPHAD-filtered candidates.

2.5.4. Property-Specific Model Configurations

Enthalpy of transformation.. ΔH was modeled using a single-output CatBoost regressor. After correlation filtering and SHAP-based elimination, a reduced feature set of six variables was retained for the final predictor.

Transformation strain.. ε_{tr} was modeled using a CatBoost regressor trained on theoretical strain responses. In addition to composition and composition-derived descriptors, inputs included loading condition (tension or compression) to capture the tension/compression asymmetry of martensitic transformation. The LDT-based theoretical-strain target and the use of λ_2 as a composition-derived descriptor follow the previously developed framework [85, 86] and are applied here without modification.

Transformation temperatures and hysteresis.. Transformation behavior was modeled using a multi-output CatBoost regressor that jointly predicted M_s , M_f , A_s , A_f , and ΔT . After the same two-stage feature selection procedure, 22 input variables were retained for the final multi-output model.

2.6. Bayesian Optimization Framework

Direct experimental evaluation of all CALPHAD-filtered candidates remained impractical. A surrogate-assisted multi-objective BO strategy was used to select compositions expected to improve the Pareto trade-off among ΔH , ϵ_{tr} , and ΔT under the additional M_s range constraint.

2.6.1. Initialization via Diverse Sampling

To ensure broad initial coverage of the screened design space, K-medoid clustering [104] was performed in the descriptor space of the 17,207 CALPHAD-filtered candidates. One representative composition from each cluster constituted the initial experimental batch (Iteration 1), emphasizing exploration and reducing the probability of seeding the campaign from a narrow region of chemistry.

2.6.2. Surrogate Modeling Across Iterations

Physics-informed CatBoost predictors provided deterministic property estimates across the candidate set. During BO iterations (Iterations 2 and 3), Gaussian process (GP) surrogate models were trained to emulate these prediction surfaces while providing uncertainty estimates required for acquisition-based decision making. Newly synthesized and measured alloys were incorporated into the training data used for subsequent model updates, enabling progressive refinement of the response surfaces in regions sampled by the campaign.

2.6.3. Acquisition Strategy and Batch Construction

Selection of new candidates was guided by expected hypervolume improvement (EHVI) [105], a standard multi-objective acquisition function. For a Pareto set $\mathcal{P}$ and reference point $\mathbf{r}$ in the objective space, the hypervolume is

$$\text{HV}(\mathcal{P}) = \lambda \left(\bigcup_{\mathbf{y} \in \mathcal{P}} [\mathbf{y}, \mathbf{r}] \right), \quad (7)$$

where $\lambda(\cdot)$ denotes the Lebesgue measure. The EHVI at a candidate $\mathbf{x}$ is

$$\text{EHVI}(\mathbf{x}) = \mathbb{E}[\text{HV}(\mathcal{P}_n \cup \{\mathbf{Y}(\mathbf{x})\}) - \text{HV}(\mathcal{P}_n)], \quad (8)$$

where $\mathcal{P}_n$ is the current Pareto front and the expectation is taken over the GP posterior for the objective vector $\mathbf{Y}(\mathbf{x})$. EHVI values were computed exactly using the algorithm of Zhao et al. [106].

Each experimental batch was constructed using the multi-scale (length-scale ensemble) framework of Couperthwaite et al. [107], which adapts the multi-scale search formulation of Joy et al. [108] to materials design. The framework addresses the strong sensitivity of GP-based acquisition predictions to the choice of kernel length-scale, particularly in high-dimensional, data-sparse settings where the maximum-likelihood estimate of the length-scale is unreliable. Rather than committing to a

single length-scale, a wide range of plausible length-scales is sampled, an independent GP is trained at each, and EHVI is evaluated over each ensemble member; the EHVI-maximizer from each ensemble member is retained as a candidate. The pooled set of candidates is then partitioned into $N = 29$ clusters by K-medoid clustering [104] in the descriptor space, and the medoid of each cluster is selected as a batch member for synthesis. Diversity within each batch emerges naturally from the clustering step, removing the need for additional diversity-enforcement heuristics. The same framework has been used in prior multi-objective alloy-discovery campaigns by the authors' group [82, 109].

In total, three experimental iterations were performed: one exploration-oriented initialization batch (Iteration 1, K-medoid seeding of the 17,207 CALPHAD-filtered candidates) followed by two BO-guided iterations (Iterations 2 and 3, EHVI maximizers across the length-scale ensemble, clustered by K-medoids). Compositions selected this way were fabricated and experimentally evaluated, and the measurements were incorporated into the training data to update the surrogate ensemble prior to the next selection cycle.

Each experimental iteration comprised a fixed-size batch limited by fabrication and characterization throughput ($N = 29$ per iteration). In MPE SMA design, a subset of compositions may show no detectable martensitic transformation under a fixed thermal protocol, either because the transformation occurs outside the accessible temperature window or because competing phases suppress transformability. Across the 87 alloys fabricated in this campaign, six showed no measurable transformation signature (7% overall), and all six occurred in the exploration-oriented first iteration (6/29, 21%). No non-transforming alloys were observed in Iterations 2 and 3 (Table 1), consistent with subsequent selection concentrating on transformation-capable regions within the constrained design space. The hit rates in Table 1 can be reproduced with a script in the public repository [89].

3. Results and Discussion

To benchmark progress toward application-relevant HTSMA performance, an operational performance window was defined to represent a desirable combination of properties. The selected thresholds, $200 \leq M_s \leq 400^\circ\text{C}$, $\Delta T \leq 50^\circ\text{C}$, $\Delta H \geq 20 \text{ J/g}$, and $\epsilon_{tr} \geq 2.5\%$, describe a challenging region of the HTSMA property space in which relatively few reported compositions simultaneously show high transformation temperatures, low thermal hysteresis, high transformation enthalpy, and large recoverable transformation strain. Although the martensite-start temperature window ($200 \leq M_s \leq 400^\circ\text{C}$) was used as a design constraint during candidate selection, the ΔT , ΔH , and ϵ_{tr} thresholds were not imposed during BO. Rather, the complete set of thresholds was introduced as an application-motivated benchmark to quantify the fraction of experimentally characterized alloys that achieved a

Table 1: Summary of alloy performance relative to the MPE HTSMA screening window ($200 \leq M_s \leq 400^\circ\text{C}$, $\Delta T \leq 50^\circ\text{C}$, $\Delta H \geq 20 \text{ J g}^{-1}$, and $\varepsilon_{\text{tr}} \geq 2.5\%$). Results are reported for the full dataset, by optimization iteration, and by elemental complexity. Percentages for the thermal properties (M_s , ΔT , and ΔH) are calculated relative to the total number of synthesized alloys in each group (N), including non-transforming alloys, demonstrating how many alloys in a set satisfied the screening criteria. Because ε_{tr} was evaluated only for a down-selected subset of alloys selected for scale-up and mechanical characterization, the reported ε_{tr} percentages represent the fraction of tested alloys satisfying the target strain criterion ($\varepsilon_{\text{tr}} \geq 2.5\%$) and are calculated using the denominator $N_{\varepsilon, \text{tested}}$. Thus, direct comparison between ε_{tr} percentages and the corresponding percentages for M_s , ΔT , and ΔH should be made with this difference in sampling population in mind.

Group	N	Non-transforming	$200 \leq M_s \leq 400$	$\Delta T \leq 20$	$\Delta T \leq 50$	$\Delta H \geq 20$	$N_{\varepsilon, \text{tested}}$	$\varepsilon_{\text{tr}} \geq 2.5^a$
Overall								
All Data	87	6 (7%)	52 (60%)	4 (5%)	57 (66%)	54 (62%)	31	24 (77%)
By Iteration								
Iteration 1	29	6 (21%)	12 (41%)	0 (0%)	15 (52%)	15 (52%)	10	7 (70%)
Iteration 2	29	0 (0%)	19 (66%)	1 (3%)	20 (69%)	16 (55%)	11	9 (82%)
Iteration 3	29	0 (0%)	21 (72%)	3 (10%)	22 (76%)	23 (79%)	10	8 (80%)
By Elemental Complexity								
Quaternary	22	0 (0%)	17 (77%)	0 (0%)	15 (68%)	18 (82%)	7	6 (86%)
Quinary	27	3 (11%)	15 (56%)	3 (11%)	18 (67%)	16 (59%)	11	10 (91%)
Senary	28	1 (4%)	18 (64%)	1 (4%)	19 (68%)	16 (57%)	11	8 (73%)
Septenary	10	2 (20%)	2 (20%)	0 (0%)	5 (50%)	4 (40%)	2	0 (0%)

^a ε_{tr} was measured only for a down-selected subset (chosen using measured M_s , ΔT , and ΔH together with the model-derived *calculated strain* metric). Therefore, ε_{tr} yields are conditional on selection for strain testing and use denominator $N_{\varepsilon, \text{tested}}$.

desirable combination of properties and to assess progress across optimization iterations. This benchmark is complementary to the BO formulation, which sought to maximize ΔH and ε_{tr} while minimizing ΔT within the targeted M_s operating window.

Because experimental measurement of ε_{tr} is substantially more time- and resource-intensive than thermal characterization and requires larger specimen sizes, ε_{tr} was evaluated only for a down-selected subset of alloys. Candidate selection was based on the measured thermal properties (M_s , ΔT , and ΔH) together with theoretical ε_{tr} calculated using Lattice Deformation Theory [85, 86]. This approach prioritized alloys selected for scale-up that were expected to show larger recoverable strains. Consequently, panel (d) of Fig. 3 reports *experimentally measured* ε_{tr} for the strain-tested subset rather than the full batch, and strain yields must use denominator $N_{\varepsilon, \text{tested}}$:

$$\text{yield}_{\varepsilon_{\text{tr}}} = \frac{N(\varepsilon_{\text{tr}} \geq 2.5\%)}{N_{\varepsilon, \text{tested}}}. \quad (9)$$

For the same reason, joint satisfaction of all four screening criteria can only be confirmed for alloys with measured ε_{tr} .

Figures 3 and 4 summarize the distributions of measured alloy properties as a function of optimization iteration and elemental complexity, respectively. Each panel combines a half-violin distribution with an overlaid boxplot, while black markers and dotted lines trace group medians. Individual alloys are plotted as right-offset swarm points to reduce overlap. Point colors encode screening status with respect to the *property shown in that panel*. Red points fall outside the panel target window. Points inside the panel

window are yellow if at least one other screening target is not satisfied *or cannot be verified* (e.g., due to unmeasured ε_{tr}); and green points indicate alloys that satisfy all four criteria simultaneously (confirmed only when ε_{tr} is available).

Beyond pass/fail counts, Fig. 3 provides a distribution-level view of campaign progression. The median M_s increased from Iteration 1 to Iteration 2 ($201.9^\circ\text{C} \rightarrow 237.9^\circ\text{C}$) and remained elevated in Iteration 3 (235.6°C), while the ΔT median decreased monotonically ($45.0^\circ\text{C} \rightarrow 37.8^\circ\text{C} \rightarrow 32.0^\circ\text{C}$). These shifts are accompanied by visibly higher point density within the shaded target bands, in agreement with the increasing yields reported in Table 1 (e.g., $12/29 \rightarrow 19/29 \rightarrow 21/29$ within $200 \leq M_s \leq 400^\circ\text{C}$ and $15/29 \rightarrow 20/29 \rightarrow 22/29$ with $\Delta T \leq 50^\circ\text{C}$). ΔH showed the clearest upward shift in Iteration 3, where the median increased to 26.6 J/g and the in-window density above $\Delta H \geq 20 \text{ J/g}$ increased (23/29; Table 1). The strain panel should be interpreted conditionally because it reflects only the strain-tested subset; accordingly, yellow points in panels (a)–(c) include cases that pass the plotted criterion but cannot be confirmed as all-four due to missing ε_{tr} .

Equally important, the prevalence of non-transforming alloys collapsed from 6/29 (21%) in the exploration-oriented Iteration 1 to 0/29 in both Iteration 2 and Iteration 3, and by Iteration 3 the great majority of alloys satisfied each of the loose screening targets individually (72% within the M_s window, 76% within $\Delta T \leq 50^\circ\text{C}$, 79% within $\Delta H \geq 20 \text{ J/g}$, and 80% of strain-tested alloys within $\varepsilon_{\text{tr}} \geq 2.5\%$) — a striking improvement given that an estimated $\sim 0.45\%$ of unconstrained MPE compositions

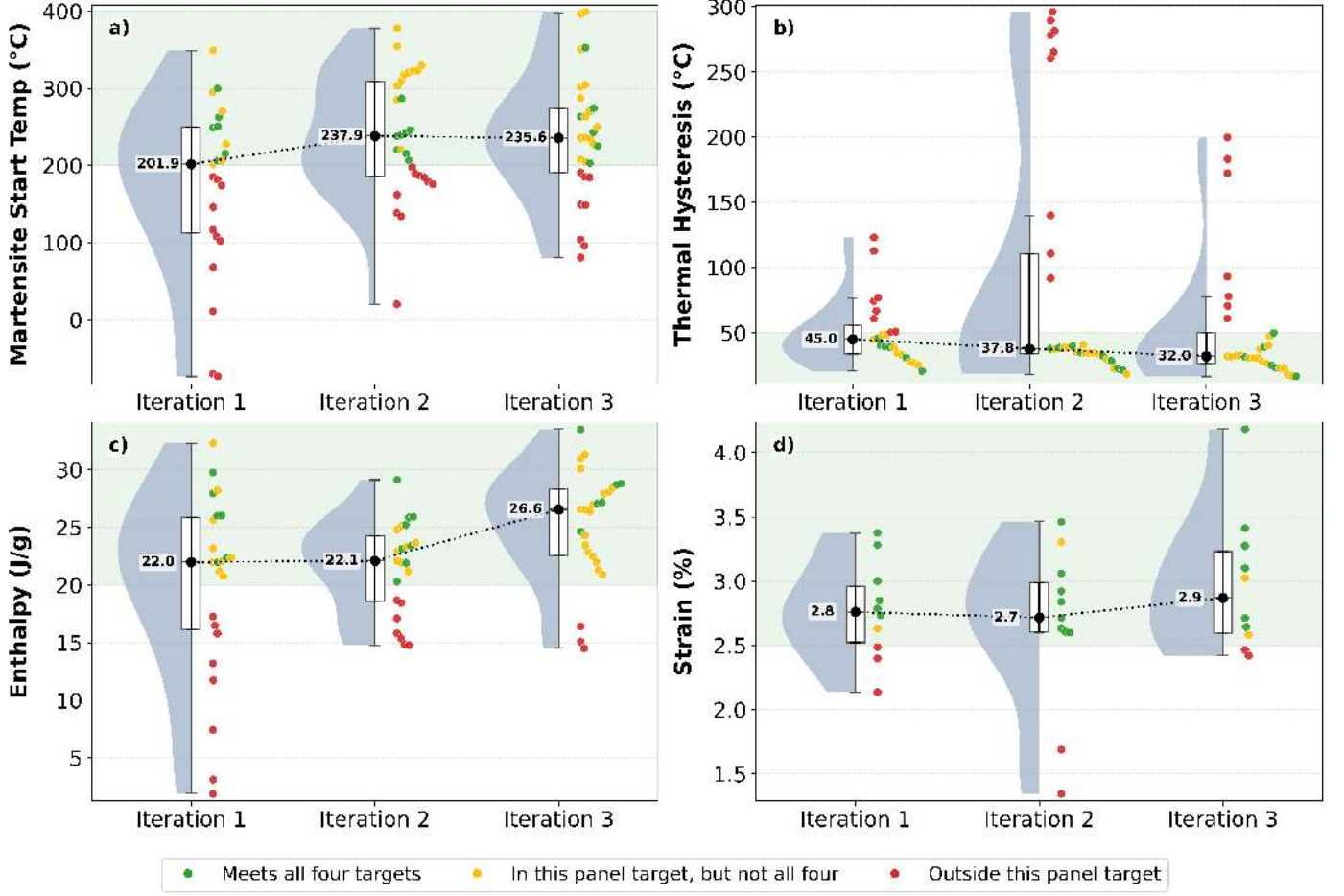


Figure 3: Distribution of alloy properties by experimental iteration: (a) martensite start temperature (M_s), (b) thermal hysteresis (ΔT), (c) transformation enthalpy (ΔH), and (d) transformation strain (ϵ_{tr}). Half-violin plots show the distribution of measured values, while the overlaid boxplots indicate the median and interquartile range. Black markers connected by dotted lines highlight the median value for each iteration. Individual alloys are shown as right-offset swarm points to visualize the underlying data and reduce overlap. Point colors indicate performance relative to the screening criteria: red points do not satisfy the target criterion for the property shown in that panel; yellow points satisfy the displayed criterion but fail at least one other screening criterion or cannot be fully evaluated because ϵ_{tr} was not measured; and green points satisfy all four screening criteria simultaneously. Note that panel (d) includes only the strain-tested subset of alloys, whereas panels (a)–(c) include all alloys with measurable transformation behavior.

pass even basic phase-stability screening and that small (<0.5 at.%) shifts in Ni/Ti suffice to move an alloy out of the transforming regime entirely.

To quantify how much of this yield is attributable to the integrated CALPHAD + ML + BO stack rather than to chance, the observed campaign yields are compared against expectations under random sampling. Combining the 0.45% CALPHAD pass rate with the all-four-pass rate observed in the exploration-oriented Iteration 1 ($6/29 \approx 21\%$ of CALPHAD-feasible alloys met all four targets) gives a baseline per-composition probability of ~ 0.001 that an arbitrary composition drawn at random from the unconstrained 3.8-million-candidate MPE space simultaneously satisfies the phase-purity, transformability, M_s , ΔT , ΔH , and ϵ_{tr} criteria. Under this baseline, 87 unconstrained random draws would yield ~ 0.087 fully qualified alloys in expectation. The campaign returned 20 confirmed all-four-pass alloys — a joint $\sim 230\times$ efficiency gain over unconstrained random sampling. The joint gain can-

not be cleanly split between CALPHAD phase-stability screening and the ML + BO framework, since the two are synergistic. This comparison is also made relative to random sampling within the predefined, literature-informed design space of approximately 3.8 million compositions, rather than an unrestricted composition space. CALPHAD narrows this space down to 17,207 alloys satisfying the $\geq 97\%$ single-phase B2 criterion — a $\sim 220\times$ reduction — but does not judge alloys on the four functional targets (M_s , ΔT , ΔH , ϵ_{tr}), and could not on its own have expanded the Pareto front. Conversely, ML + BO without a CALPHAD prior would be highly inefficient in this alloy family. The algorithm would repeatedly recommend compositions that are not single-phase B2 under the fixed $950^\circ\text{C} / 24\text{h}$ homogenization schedule, wasting most of the experimental budget on materials that never make it past phase-stability.

Observationally, the fraction of alloys meeting all four loose target thresholds remains similar in the K-medoid-

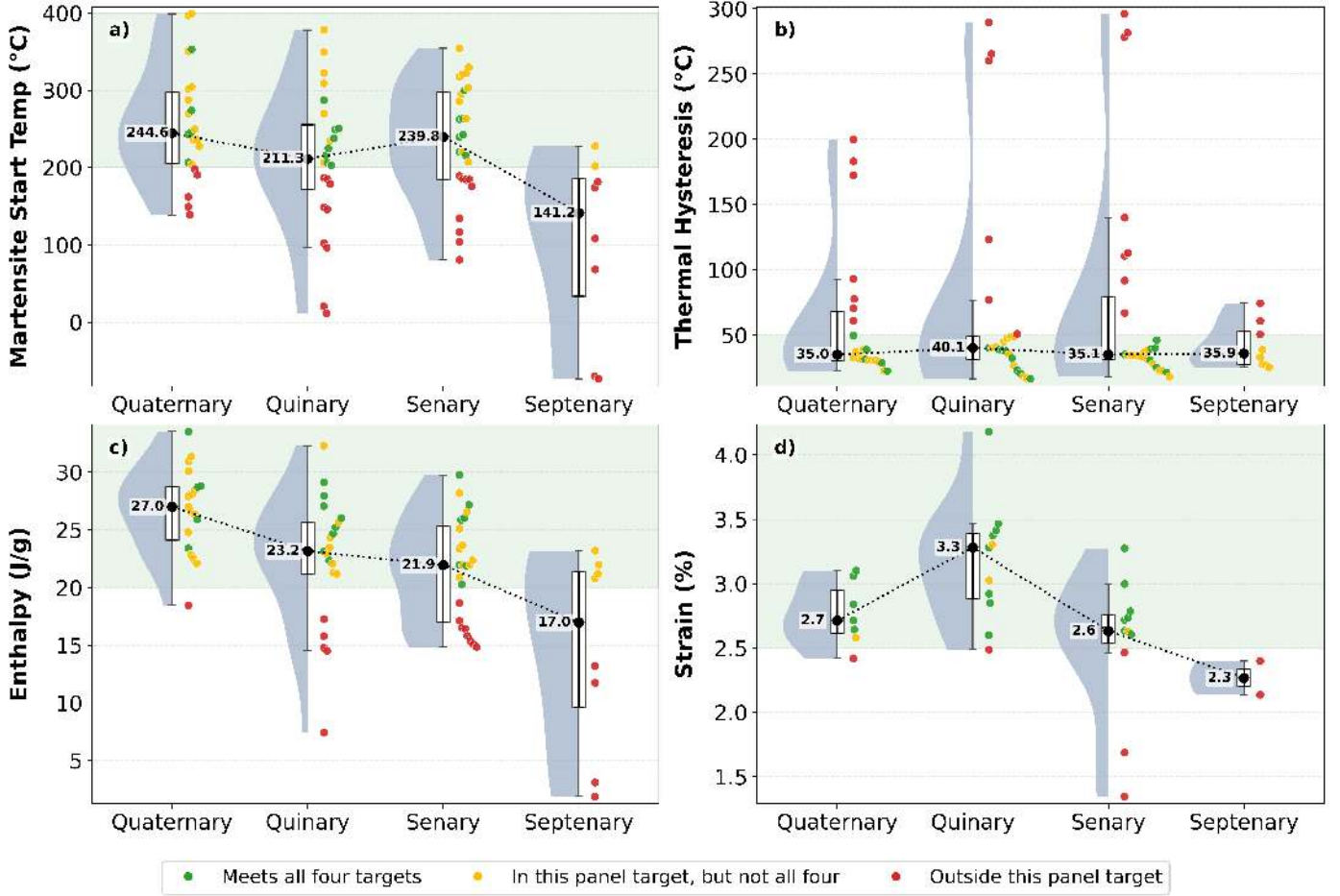


Figure 4: Property distributions as a function of elemental complexity, from quaternary to septenary compositions. The figure uses the same plotting conventions as Fig. 3. Half-violin plots show the distribution of measured values, while the overlaid boxplots indicate the median and interquartile range. Black markers connected by dotted lines highlight the median value for each complexity level. Individual alloys are shown as right-offset swarm points to visualize the underlying data and reduce overlap. Point colors indicate performance relative to the screening criteria. Red points correspond to alloys that do not satisfy the target criterion for the property shown in that panel. Yellow points satisfy the displayed criterion but fail at least one other screening criterion or cannot be fully evaluated because ε_{tr} was not measured. Green points represent alloys that satisfy all four screening criteria simultaneously. As in Fig. 3, M_s , ΔT , and ΔH include all alloys within each complexity class, whereas ε_{tr} reflects only the strain-tested subset selected for mechanical characterization.

seeded Iteration 1 ($6/29 = 21\%$) and in the two BO-guided iterations combined ($14/58 = 24\%$), so BO does not materially improve the loose-threshold hit rate within the CALPHAD-feasible subset. This metric, however, misses the more relevant markers of BO’s contribution — Pareto-front expansion, the collapse of non-transforming compositions to 0/58 in Iterations 2–3, and iteration-to-iteration improvement of median properties. The $\sim 230\times$ joint gain is therefore a synergistic outcome — CALPHAD as a physics-informed filter that removes chemically infeasible compositions, and multi-objective BO as a targeted search within the feasible subset that finds alloys in previously unexplored regions of the property space. This is the case for metallurgy-informed rather than prior-agnostic alloy design [110]. The derivation of these baseline probabilities, the Poisson-tail computation, and the robustness analysis at 95% confidence-interval bounds of the conditional rate are given in Appendix A.

The closed-loop ML + BO contribution is most evi-

dent in the property measured on every synthesized alloy: transformability. In Iteration 1 (K-medoid-seeded, no BO guidance), 6 of 29 alloys were non-transforming (21%), consistent with the difficulty of avoiding non-transforming compositions within CALPHAD-feasible space given that shifts of less than 0.5 at.% in Ni/Ti suffice to move an otherwise phase-pure alloy from transforming to non-transforming. By Iterations 2 and 3 the non-transforming rate fell to 0 out of 58 alloys. If the 21% Iteration 1 rate is taken as the CALPHAD-only baseline, the probability of observing zero non-transforming alloys in 58 independent CALPHAD-feasible draws is $(1 - 0.21)^{58} \approx 1.2 \times 10^{-6}$, and remains below 0.3% even at the lower bound of a 95% confidence interval on the 21% Iteration 1 rate (Appendix A). The closed-loop ML + BO predictor therefore reliably concentrates the search within the transformation-capable subregions of CALPHAD-feasible space across 58 consecutive synthesized alloys — a substantive result given the steep composition sensitivity of the martensitic trans-

formation in this alloy family — and complements its observed contributions to the median ΔT , median ΔH , and Pareto-front expansion reported earlier.

A complementary BO-merit metric — per-alloy Pareto-front membership — shows the same pattern. Of the 31 strain-tested alloys (10/11/10 from Iterations 1/2/3), 10 lie on the final cumulative Pareto front of the $(-\Delta T, \Delta H, \varepsilon_{\text{tr}})$ objective space (here $\Delta T = A_{50} - M_{50}$ from the second DSC cycle, ΔH the second-cycle average enthalpy, and ε_{tr} the largest measured transformation strain), contributed in a 2/3/5 split across the three iterations. The per-alloy Pareto-membership rate therefore rises from 20% in Iteration 1 (no BO guidance) to 27% in Iteration 2 and 50% in Iteration 3 — a $2.5\times$ lift from Iteration 1 to Iteration 3. Under a random-sampling baseline calibrated on the Iteration 1 rate, the probability of producing ≥ 5 Pareto-optimal alloys in 10 Iteration 3 strain-tested draws is $\sim 3.3\%$ (Appendix A). Iteration 3 also *displaces* 3 earlier Pareto points: of the 8 Pareto-optimal alloys after Iteration 2 (3 from Iteration 1, 5 from Iteration 2), 3 are strictly dominated by Iteration 3 alloys and removed from the cumulative front. The closed-loop BO is therefore not only expanding the Pareto frontier with additional points but actively refining it by identifying compositions that dominate previously Pareto-optimal alloys.

Beyond these iteration-level metrics, the campaign identified $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ as the highest-performing composition — the lowest measured ΔT and the highest measured ε_{tr} in the entire campaign — and, as Fig. 9 shows, this composition outperforms every previously reported bulk HTSMA in the $200 \leq M_s \leq 400^\circ\text{C}$ window on the combined $(\Delta T, \varepsilon_{\text{tr}})$ objectives. This performance is achieved at only 2 at.% Pd, far below the precious-metal fractions typical of conventional NiTiPd HTSMAs. Finding such a low-Pd composition that is also Pareto-optimal in the $(\Delta T, \varepsilon_{\text{tr}})$ space within three closed-loop iterations demonstrates the value of the ML + BO framework, given the steep dependence of ΔT and ε_{tr} on small composition variations within this alloy family.

Figure 4 complements the iteration trends by isolating how the response distributions changed with elemental complexity. In principle, higher complexity introduces additional compositional degrees of freedom that can enable combinatorial benefits (e.g., tuning M_s or increasing ε_{tr}). In the present campaign, however, increasing complexity correlated most strongly with a penalty in ΔH : the median ΔH decreased systematically from quaternary to septenary compositions ($27.0 \rightarrow 23.2 \rightarrow 22.0 \rightarrow 17.0$ J/g), which aligned with the reduced fraction of septenary alloys meeting $\Delta H \geq 20$ J/g (Table 1). Septenary alloys also showed a substantially lower median M_s (141.2°C) and a visibly larger red population in the M_s panel, as reflected in only 2/10 satisfying $200 \leq M_s \leq 400^\circ\text{C}$. The trends observed for septenary alloys should be considered preliminary because this group includes only 10 alloys, of which only two were tested for transformation strain. These lim-

ited measurements do not establish a general relationship between increasing compositional complexity and functional performance, particularly recoverable transformation strain.

By contrast, ε_{tr} did not show a monotonic complexity trend in this dataset: within the strain-tested subset, quinary alloys had the highest median ε_{tr} (3.3%) relative to quaternary (2.7%), senary (2.6%), and septenary (2.3%). However, this comparison is conditional on targeted strain measurements rather than unconditional yields across all synthesized alloys. The ΔH reduction at high complexity provided a plausible mechanism for the optimization campaign drifting toward simpler chemistries: the mean elemental complexity of experimentally selected alloys decreased from 5.93 elements/alloy in Iteration 1 to 4.62 elements/alloy in Iteration 3. Thus, the selection strategy balanced the multi-objective trade-offs by favoring regions with higher ΔH while maintaining acceptable M_s and ΔT .

Complementing the distribution plots, Table 3 lists the property ranges for alloys in each class (satisfying, partially satisfying, or failing the target window). For alloys satisfying all four performance benchmarks (green class), the observed property ranges are $M_s = 203.1\text{--}352.9^\circ\text{C}$, $\Delta T = 16.5\text{--}49.8^\circ\text{C}$, $\Delta H = 20.29\text{--}33.52$ J/g, and $\varepsilon_{\text{tr}} = 2.60\text{--}4.18\%$.

Yellow points often extend closer to target-window edges (e.g., M_s up to 398.9°C or ΔT down to 16.9°C), highlighting alloys that satisfy the displayed criterion but fail (or cannot be confirmed to satisfy) at least one other requirement. Conversely, the red ranges quantify how far out-of-window failures extend (e.g., ΔT up to 296°C and ΔH down to 1.88 J/g), while also emphasizing that N_{avail} can differ across properties and groups due to non-transforming cases and selective strain follow-up.

While Table 3 summarizes the achievable property ranges, Table 4 identifies the specific compositions associated with the extreme values in each color class. These compositions define the practical performance limits observed in the current dataset and highlight alloys occupying particularly favorable or unfavorable regions of the multi-objective design space.

In this study, the optimization priorities are explicit: maximize recoverable ε_{tr} and ΔH , while minimizing ΔT , under the constraint that M_s remains within the operational window. Table 4 complements the distribution plots by anchoring each color class with extremal compositions and their associated values. Extremal compositions within each color class define the performance envelope and identify candidates that achieve an exceptional value in one objective while remaining competitive on the others.

One of the most notable compositions identified in this study is $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ (Fig. 5), which anchors the green (all-four) set. Among the alloys satisfying all four screening criteria, it shows both the lowest thermal hysteresis observed ($\Delta T = 16.5^\circ\text{C}$) and the highest measured transformation strain in that set ($\varepsilon_{\text{tr}} = 4.18\%$). This com-

Table 2: Summary of BO-effectiveness metrics narrated in Section 3 with null-hypothesis tail probabilities. Baselines are calibrated from the K-medoid-seeded Iteration 1; full derivations and 95% Wilson-CI robustness ranges are given in Appendix A.

Metric	Iter 1 (calibration)	BO iterations	Null-model tail probability
Non-transforming rate	6/29 (21%)	0/58 (Iter 2+3)	$\sim 1.2 \times 10^{-6}$ (§A.5)
Joint 4-pass rate	6/29 (21%)	14/58 (24%)	joint pipeline $\sim 230\times$ (§A.3–A.4)
Per-alloy Pareto membership (strain-tested)	2/10 (20%)	5/10 (Iter 3)	~ 0.033 (§A.6)

Table 3: Summary of property ranges associated with the color-coded performance categories used in Figs. 3 and 4. For each group, N_{avail} denotes the number of alloys with an available measurement for the corresponding property; non-transforming alloys and unmeasured entries are excluded. Green points represent alloys that satisfy all four performance benchmarks, yellow points satisfy the benchmark for the displayed property but fail at least one other benchmark (or cannot be fully evaluated because ε_{tr} was not measured), and red points fall outside the benchmark range for the displayed property. Percentages are calculated within each property/group combination using $N_{\text{avail}} = N_{\text{avail}}(\text{green}) + N_{\text{avail}}(\text{yellow}) + N_{\text{avail}}(\text{red})$. N_{avail} can be smaller than the total number of synthesized alloys because some compositions exhibited no measurable martensitic transformation and because ε_{tr} was evaluated only for the strain-tested subset. Reported values correspond to the min. and max. observed within each color class.

Property	Group	N_{avail}	Green: all four	Yellow: panel-only	Red: outside panel
Martensite start temperature					
M_s ($^{\circ}\text{C}$)	Whole Data	81	20 (25%), 203.1–352.9	32 (40%), 201.9–398.9	29 (36%), –73.3–197.8
M_s ($^{\circ}\text{C}$)	Iteration 1	23	6 (26%), 205.6–299.7	6 (26%), 201.9–349.0	11 (48%), –73.3–185.2
M_s ($^{\circ}\text{C}$)	Iteration 2	29	8 (28%), 206.6–286.9	11 (38%), 220.2–378.1	10 (34%), 20.3–197.8
M_s ($^{\circ}\text{C}$)	Iteration 3	29	6 (21%), 203.1–352.9	15 (52%), 204.2–398.9	8 (28%), 80.7–190.7
M_s ($^{\circ}\text{C}$)	Quaternary	22	5 (23%), 206.6–352.9	12 (55%), 204.2–398.9	5 (23%), 138.6–197.8
M_s ($^{\circ}\text{C}$)	Quinary	24	8 (33%), 203.1–286.9	7 (29%), 206.7–378.1	9 (38%), 11.3–186.5
M_s ($^{\circ}\text{C}$)	Senary	27	7 (26%), 215.6–299.7	11 (41%), 207.3–354.3	9 (33%), 80.7–189.2
M_s ($^{\circ}\text{C}$)	Septenary	8	0 (0%), –	2 (25%), 201.9–227.7	6 (75%), –73.3–181.5
Thermal hysteresis					
ΔT ($^{\circ}\text{C}$)	Whole Data	81	20 (25%), 16.5–49.8	37 (46%), 16.9–48.9	24 (30%), 50.5–296.0
ΔT ($^{\circ}\text{C}$)	Iteration 1	23	6 (26%), 20.6–46.1	9 (39%), 25.4–48.9	8 (35%), 50.5–123.2
ΔT ($^{\circ}\text{C}$)	Iteration 2	29	8 (28%), 21.5–39.8	12 (41%), 18.1–40.8	9 (31%), 91.7–296.0
ΔT ($^{\circ}\text{C}$)	Iteration 3	29	6 (21%), 16.5–49.8	16 (55%), 16.9–47.4	7 (24%), 60.8–199.9
ΔT ($^{\circ}\text{C}$)	Quaternary	22	5 (23%), 22.4–49.8	10 (45%), 22.7–38.7	7 (32%), 60.8–199.9
ΔT ($^{\circ}\text{C}$)	Quinary	24	8 (33%), 16.5–40.0	10 (42%), 16.9–48.9	6 (25%), 50.9–289.4
ΔT ($^{\circ}\text{C}$)	Senary	27	7 (26%), 21.5–46.1	12 (44%), 18.1–38.7	8 (30%), 66.9–296.0
ΔT ($^{\circ}\text{C}$)	Septenary	8	0 (0%), –	5 (62%), 25.4–38.7	3 (38%), 50.5–74.3
Transformation enthalpy					
ΔH (J/g)	Whole Data	72	20 (28%), 20.29–33.52	34 (47%), 20.77–32.29	18 (25%), 1.88–18.88
ΔH (J/g)	Iteration 1	23	6 (26%), 21.97–29.75	9 (39%), 20.77–32.29	8 (35%), 1.88–17.27
ΔH (J/g)	Iteration 2	23	8 (35%), 20.29–29.13	8 (35%), 21.17–25.38	7 (30%), 14.66–18.88
ΔH (J/g)	Iteration 3	26	6 (23%), 24.67–33.52	17 (65%), 20.89–31.34	3 (12%), 14.51–16.43
ΔH (J/g)	Quaternary	19	5 (26%), 23.41–33.52	13 (68%), 22.10–31.34	1 (5%), 18.88–18.88
ΔH (J/g)	Quinary	21	8 (38%), 22.33–29.13	8 (38%), 21.17–32.29	5 (24%), 7.41–17.27
ΔH (J/g)	Senary	24	7 (29%), 20.29–29.75	9 (38%), 20.89–28.20	8 (33%), 14.66–18.06
ΔH (J/g)	Septenary	8	0 (0%), –	4 (50%), 20.77–23.19	4 (50%), 1.88–13.19
Transformation strain					
ε_{tr} (%)	Whole Data	31	20 (65%), 2.60–4.18	4 (13%), 2.58–3.30	7 (23%), 1.35–2.49
ε_{tr} (%)	Iteration 1	10	6 (60%), 2.73–3.38	1 (10%), 2.63–2.63	3 (30%), 2.14–2.49
ε_{tr} (%)	Iteration 2	11	8 (73%), 2.60–3.46	1 (9%), 3.30–3.30	2 (18%), 1.35–1.69
ε_{tr} (%)	Iteration 3	10	6 (60%), 2.64–4.18	2 (20%), 2.58–3.03	2 (20%), 2.42–2.46
ε_{tr} (%)	Quaternary	7	5 (71%), 2.64–3.10	1 (14%), 2.58–2.58	1 (14%), 2.42–2.42
ε_{tr} (%)	Quinary	11	8 (73%), 2.60–4.18	2 (18%), 3.03–3.30	1 (9%), 2.49–2.49
ε_{tr} (%)	Senary	11	7 (64%), 2.61–3.27	1 (9%), 2.63–2.63	3 (27%), 1.35–2.46
ε_{tr} (%)	Septenary	2	0 (0%), –	0 (0%), –	2 (100%), 2.14–2.40

Percentages are computed within each (property, group) using $N_{\text{avail}} = \text{green} + \text{yellow} + \text{red}$. N_{avail} can be smaller than the batch size due to non-transforming alloys lacking extractable $M_s/\Delta T$ or due to selective follow-up testing for ε_{tr} .

bination makes it a strong candidate for high-temperature actuation, where reduced ΔT losses are required without sacrificing actuation amplitude. The composition is also practically noteworthy because these properties are achieved with only 2 at.% Pd. Although Pd-containing NiTi-based SMAs can be cost-prohibitive, this result suggests that relatively small Pd additions may be sufficient to substantially reduce thermal hysteresis without requiring the high Pd fractions commonly used in NiTiPd alloys to reach elevated transformation temperatures. More

broadly, the result suggests that Pd may act as an effective modifier of transformation hysteresis, while the primary increase in transformation temperature is supplied by Ti-site substitution through Hf additions.

Pd appears to exert an outsized influence on hysteresis at low concentration, enabling a low-loss transformation pathway in a Hf-containing matrix while keeping precious-metal usage minimal. A second practical implication is that this composition helps separate two roles that are often conflated in Pd-bearing NiTi systems: Pd can be

used as a *hysteresis-control lever* even when the dominant transformation-temperature raising is supplied by Ti-site substitution (here, Hf). A candidate mechanism, consistent with the crystallographic compatibility calculation reported below and potentially testable in future atomistic work, is that Pd on the Ni sublattice modifies the B2 $\rightarrow$ B19' transformation stretch such that λ_2 is driven closer to unity. The measured lattice parameters yield $\lambda_2 = 0.9449$ for the 2 at.% Pd alloy versus $\lambda_2 = 0.9325$ for the Pd-free lowest- ΔT alloy. Under the Ball–James geometric-compatibility criterion, $\lambda_2 \rightarrow 1$ lowers the elastic energy stored at habit-plane interfaces and would reduce the nucleation barrier for the reverse transformation, offering a potential explanation for the hysteresis reduction without requiring the high Pd fractions used in ternary Ni–TiPd. Across the campaign alloys, λ_2 closer to unity is associated with lower hysteresis (see below). Systematic Pd-content variation at fixed base composition, DFT prediction of the composition-dependent B2 and B19' lattice parameters, and coupled in-situ characterization would confirm this.

Very low ΔT values are also accessible outside the all-four subset: the yellow minimum ($\Delta T = 16.9^\circ\text{C}$, $\text{Ni}_{43.2}\text{Ti}_{24.8}\text{Co}_4\text{Pd}_3\text{Hf}_{25}$) is only marginally above the green minimum. The proximity of these minima indicates that the alloy design space contains a local low-hysteresis basin spanning $\sim 2\text{--}3$ at.% Pd, where small variations in Co content and base Ni/Ti balance can preserve low ΔT . This suggests a *robust* low- ΔT neighborhood rather than an isolated point solution, which increases the likelihood that subsequent processing refinements or minor chemistry adjustments could improve secondary objectives without sacrificing ΔT . Small numeric differences between alloys near the classification thresholds fall within the measurement uncertainties documented in the inter-laboratory studies underlying the tangent-intercept methods of ASTM F2004 (DSC) and ASTM E3097 (UCFTC). The iteration-to-iteration property trends — median ΔT from 45.0 to 32.0°C and median ΔH from 22.0 to 26.6 J/g between Iterations 1 and 3 — are much larger than any plausible measurement uncertainty, so the campaign-level Pareto-front and hypervolume conclusions are robust, although the Pareto membership of individual closely spaced alloys may be sensitive to it.

At the same time, the shift from green to yellow reflects the multi-objective constraint: achieving low ΔT alone is insufficient if other requirements (e.g., the M_s window, ΔH threshold, or confirmed ε_{tr}) are not simultaneously satisfied or cannot be verified due to limited strain measurements. This motivates targeted follow-up within the same chemistry neighborhood to determine whether additional members can be “promoted” from panel-only low ΔT (yellow) to globally qualified behavior (green) as more strain measurements become available.

On the energy side, the green class spans $\Delta H = 20.29\text{--}33.52$ J/g. The lowest-enthalpy alloy that still meets all four criteria is $\text{Ni}_{43}\text{Ti}_{24}\text{Cu}_5\text{Co}_2\text{Hf}_{20}\text{Zr}_6$ ($\Delta H =$

20.29 J/g), indicating that globally qualified compositions can sit close to the ΔH threshold and still satisfy the full screening window. In contrast, the high-enthalpy extreme $\text{Ni}_{49}\text{Ti}_{25}\text{Co}_1\text{Hf}_{25}$ reaches $\Delta H = 33.52$ J/g but also sits at the upper end of green-class thermal hysteresis ($\Delta T = 49.8^\circ\text{C}$), illustrating that pushing latent heat upward can consume ΔT margin even within the qualified set. These extremes imply that the most informative candidates are those that maintain *headroom* against constraints while offering strong multi-objective balance. $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ combines the best-in-campaign ΔT within the all-four set with a position near the favorable end of the strain envelope, making it a plausible starting point for subsequent optimization aimed at increasing ΔH without forfeiting low ΔT .

In addition to reporting the fraction of compositions that satisfy the operational screening window, multi-objective progress is quantified using the (normalized) hypervolume (HV) indicator (Fig. 6). HV measured the volume of objective space dominated by the current non-dominated set (Pareto-front approximation) relative to a fixed reference point. ΔT was treated as a minimization objective by transforming to an all-maximization space via $(-\Delta T, \Delta H, \varepsilon_{\text{tr}})$. To prevent differences in units and dynamic range from dominating HV, each objective is min-max normalized using pooled statistics over all available data,

$$\tilde{\mathbf{y}} = \frac{\mathbf{y} - \mathbf{y}_{\min}}{\mathbf{y}_{\max} - \mathbf{y}_{\min}}. \quad (10)$$

where $\mathbf{y}_{\min}$ and $\mathbf{y}_{\max}$ are taken component-wise in the transformed all-maximization space. HV was computed cumulatively (Iteration 1; Iterations 1–2 pooled; Iterations 1–3 pooled) and was evaluated only for alloys with all three objective values available (i.e., jointly measured ΔT , ΔH , and ε_{tr}).

The reference point was defined as a conservative *worst-observed* composite coordinate: the component-wise pooled minimum in the transformed space, shifted by a small margin to ensure it is strictly dominated by the evaluated solutions,

$$\mathbf{r} = \mathbf{y}_{\min} - \epsilon (\mathbf{y}_{\max} - \mathbf{y}_{\min}), \quad (11)$$

with $\epsilon = 0.05$ in this work. Under this definition, increases in HV indicated that the cumulative non-dominated set expanded toward simultaneously lower ΔT and higher ΔH and strain.

Figure 6a showed that cumulative HV increased from 0.544 (Iteration 1) to 0.577 (Iteration 2) and then rose sharply to 0.808 (Iteration 3). The corresponding gains (Fig. 6b) were modest from Iteration 1 to 2 (6.0%) but substantially larger from Iteration 2 to 3 (40.1%). Because HV is sensitive to both the appearance of new non-dominated points and outward movement of the existing frontier, the Iteration 3 increase was consistent with a marked expansion of the achievable trade-off envelope across the three objectives. Cumulatively, this corresponds to a 48.5% ex-

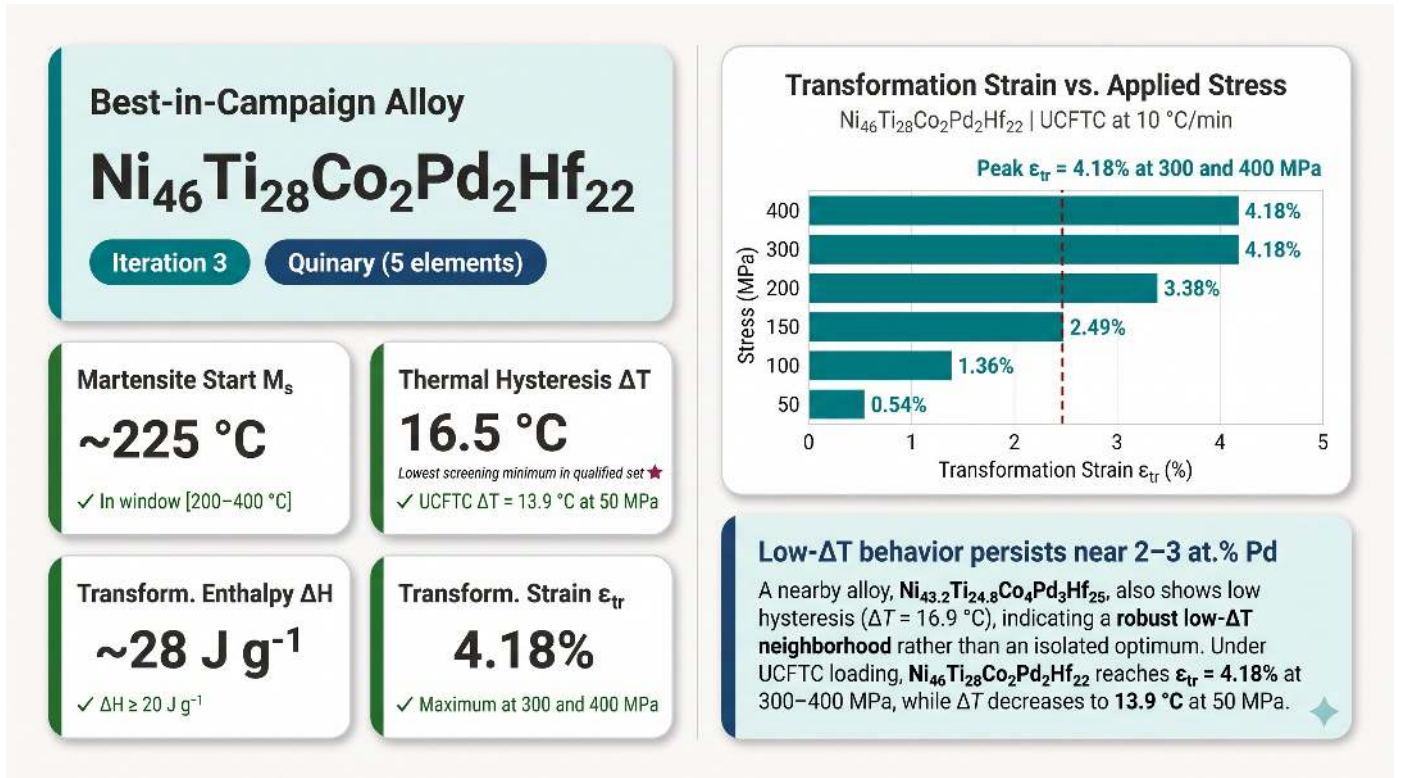


Figure 5: Summary of the highest-performing alloy identified during the Bayesian optimization campaign, $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ (Iteration 3). The alloy satisfies all four performance benchmarks and exhibits the lowest thermal hysteresis ($\Delta T = 16.5^\circ\text{C}$) and highest measured transformation strain ($\epsilon_{\text{tr}} = 4.18\%$) among the fully qualified compositions. Property panels summarize the key screening metrics, including martensite start temperature (M_s), thermal hysteresis (ΔT), transformation enthalpy (ΔH), and transformation strain (ϵ_{tr}). The bar chart shows the evolution of ϵ_{tr} with applied stress during uniaxial constant-force thermal cycling (UCFTC). The figure also highlights the persistence of low-hysteresis behavior in nearby Pd-containing compositions and the reduction of ΔT under applied load. Figure generated using Gemini and subsequently reviewed and edited by the authors.

Table 4: Minimum and maximum property values observed within each color category for the full dataset (“Whole Data”), together with the corresponding alloy compositions. The color categories are consistent with those used in Figs. 3 and 4. Green entries represent alloys that satisfy all four performance benchmarks (confirmed only for alloys with measured ϵ_{tr}), yellow entries satisfy the benchmark for the displayed property but fail at least one other benchmark or cannot be fully evaluated because ϵ_{tr} was not measured, and red entries fall outside the benchmark range for the displayed property. The table identifies the compositions defining the observed property limits within each category and highlights alloys occupying favorable or unfavorable regions of the multi-objective design space. A complete list of all compositions generated, synthesized, processed, and characterized during the high-throughput campaign is provided in the Supplementary Information.

Property	Green: min–max (composition)	Yellow: min–max (composition)	Red: min–max (composition)
Martensite start temperature			
M_s ($^\circ\text{C}$)	203.1 ($\text{Ni}_{47}\text{Ti}_{27}\text{Co}_3\text{Hf}_{12}\text{Zr}_{11}$) 352.9 ($\text{Ni}_{49}\text{Ti}_{25}\text{Co}_1\text{Hf}_{25}$)	201.9 ($\text{Ni}_{36}\text{Ti}_{23}\text{Cu}_5\text{Co}_5\text{Pd}_4\text{Hf}_{21}\text{Zr}_6$) 398.9 ($\text{Ni}_{50}\text{Ti}_{23}\text{Hf}_{10}\text{Zr}_{17}$)	−73.3 ($\text{Ni}_{35}\text{Ti}_{24}\text{Cu}_5\text{Co}_3\text{Pd}_7\text{Hf}_{19}\text{Zr}_7$) 197.8 ($\text{Ni}_{50}\text{Ti}_{28}\text{Hf}_6\text{Zr}_{16}$)
Thermal hysteresis			
ΔT ($^\circ\text{C}$)	16.5 ($\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$) 49.8 ($\text{Ni}_{49}\text{Ti}_{25}\text{Co}_1\text{Hf}_{25}$)	16.9 ($\text{Ni}_{43.2}\text{Ti}_{24.8}\text{Co}_4\text{Pd}_3\text{Hf}_{25}$) 48.9 ($\text{Ni}_{42}\text{Ti}_{25}\text{Co}_2\text{Pd}_6\text{Hf}_{25}$)	50.5 ($\text{Ni}_{35}\text{Ti}_{22}\text{Cu}_5\text{Co}_3\text{Pd}_7\text{Hf}_{20}\text{Zr}_8$) 296.0 ($\text{Ni}_{46}\text{Ti}_{20}\text{Co}_3\text{Pd}_1\text{Hf}_{15}\text{Zr}_{15}$)
Transformation enthalpy			
ΔH (J/g)	20.29 ($\text{Ni}_{43}\text{Ti}_{24}\text{Cu}_5\text{Co}_2\text{Hf}_{20}\text{Zr}_6$) 33.52 ($\text{Ni}_{49}\text{Ti}_{25}\text{Co}_1\text{Hf}_{25}$)	20.77 ($\text{Ni}_{39}\text{Ti}_{24}\text{Cu}_5\text{Co}_5\text{Pd}_1\text{Hf}_{18}\text{Zr}_8$) 32.29 ($\text{Ni}_{49}\text{Ti}_{25}\text{Pd}_1\text{Hf}_{20}\text{Zr}_5$)	1.88 ($\text{Ni}_{36}\text{Ti}_{23}\text{Cu}_5\text{Co}_2\text{Pd}_7\text{Hf}_{19}\text{Zr}_8$) 18.88 ($\text{Ni}_{45.2}\text{Ti}_{25.8}\text{Co}_5\text{Hf}_{24}$)
Transformation strain			
ϵ_{tr} (%)	2.60 ($\text{Ni}_{48}\text{Ti}_{26}\text{Co}_2\text{Hf}_8\text{Zr}_{16}$) 4.18 ($\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$)	2.58 ($\text{Ni}_{50}\text{Ti}_{27}\text{Hf}_{18}\text{Zr}_5$) 3.30 ($\text{Ni}_{45}\text{Ti}_{25}\text{Co}_5\text{Hf}_{11}\text{Zr}_{14}$)	1.35 ($\text{Ni}_{46}\text{Ti}_{22}\text{Co}_3\text{Pd}_1\text{Hf}_{21}\text{Zr}_7$) 2.49 ($\text{Ni}_{49}\text{Ti}_{25}\text{Pd}_1\text{Hf}_{20}\text{Zr}_5$)

pansion of the accessible trade-off envelope from Iteration 1 to Iteration 3.

Taken together with the monotonic rise in the screening-window yield reported in Table 1, this hypervolume progression meets the falsifiable expectation set out in Section 2.5: when the CALPHAD + ML + BO stack operates as intended, the closed-loop search becomes progressively more efficient at producing alloys against the screening targets and at expanding the achievable Pareto front. The iteration-over-iteration efficiency gain is the operational signal that the surrogate models — used here largely as fixed filters, with the transformation-temperature and ΔH CatBoost predictors incrementally retrained on appended MPE measurements — preserve rank-ordering among candidates well enough for the acquisition function to make useful selections. The accuracy of the surrogates on the higher-dimensional MPE chemistry can be assessed post-hoc against the 81 transforming alloys synthesized during the campaign, which serve as an implicit hold-out for models trained primarily on binary-quaternary NiTi data. Two complementary metrics are relevant. Because the goal in BO is to find the argument that optimizes a function rather than to predict the value of the function at every point, a model that is good at ranking remains useful even when its pointwise accuracy is limited. Spearman’s ρ measures rank-order preservation, and mean absolute error (MAE) measures pointwise accuracy.

By the rank-order metric, the surrogates transfer to the MPE chemistry reasonably well. The Spearman rank correlations between predicted and measured values (Table 5) are positive and statistically significant for the aggregate dataset across all four properties ($p < 10^{-4}$). Each correlation is higher in Iteration 3 than in Iteration 1. For M_s , ρ rises from 0.27 in Iteration 1 (no BO guidance) to 0.73 in Iteration 3.

The overall MAEs, in contrast, are larger than on the original NiTi datasets, as expected for compositions outside the training domain: 64.3 °C for M_s and 87.1 °C for A_f , compared with a hold-out MAE of 17.4 °C for the original transformation-temperature model, and 7.3 J/g for ΔH . In-loop retraining recovers a substantial fraction of this. MAE M_s drops from 83.9 °C in Iteration 1 to 43.1 °C in Iteration 3, and MAE ΔH from 11.6 J/g to 4.5 J/g. Because BO relies on the relative ranking of candidates, this level of rank-order agreement is sufficient to guide the acquisition.

The λ_2 and LDT models, which were used as fixed filters, can be checked in the same way against the 70 synthesized alloys whose B2 and B19’ lattice parameters were measured. The λ_2 values computed from these lattice parameters agree with the composition-based λ_2 model with a Spearman ρ of 0.49 ($p < 10^{-4}$) and a mean absolute error of 0.012, close to the error of 0.01 reported on the original NiTi data [85]. Applied to the predicted λ_2 , the LDT model reproduces the dependence of the theoretical transformation strain on loading direction obtained from the measured lattice parameters ($\rho = 0.92$ over twelve direc-

tions in tension and compression), and all 70 alloys exceed the 2.5% screening threshold in every direction, both by the model and by the measured lattice parameters. The data and scripts for these comparisons and for Table 5 are available in the public repository [89].

The composition-only surrogates do not directly capture several MPE-specific factors — secondary phases such as Ti₂Ni (Figure 7), local chemical environments and short-range order effects that composition-averaged descriptors cannot resolve, precipitation kinetics at intermediate temperatures, and coupled cyclic effects such as functional fatigue. The models here function as rank-order priors for the BO acquisition rather than pointwise predictors of MPE behavior.

Figure 7 shows representative microstructures for three alloys with varied chemical complexity from quaternary to septenary. The microstructure was predominantly a light-contrast matrix containing a dark-contrast secondary phase uniformly distributed with no strong evidence of clustering. This two-phase morphology was representative of all alloys processed under the present arc-melting/solution-treatment route. The phase was identified as Ti₂Ni via XRD, consistent with previous observations [111]. Image analysis of the micrographs in Figure 7 gives Ti₂Ni-type area fractions of 1.1%, 1.3%, and 1.6% for Ni₃₆Ti₂₃Cu₅Co₂Pd₇Hf₁₉Zr₈, Ni₄₁Ti₂₅Cu₆Co₃Hf₁₈Zr₇, and Ni₄₉Ti₂₉Co₁Hf₂₁, respectively. The matrix therefore makes up more than 98% of the microstructure in all three alloys, consistent with the > 97% single-phase B2 criterion used in the CALPHAD screening. The measured room-temperature area fraction does not, however, directly validate the predicted equilibrium phase fraction at 950 °C, because the processed microstructure also reflects solidification history, homogenization kinetics, and impurity effects. Residual Ti₂Ni-type particles may originate from local compositional segregation during solidification and persist if homogenization does not fully dissolve them, and oxygen incorporation can additionally stabilize isostructural Ti₄Ni₂O_x-type constituents [112, 113]. Particle oxygen content and formation history were not established in this study, so these remain plausible explanations rather than demonstrated mechanisms. The CALPHAD criterion is accordingly an equilibrium composition-screening condition rather than a guarantee of the final microstructure. The $\Delta H \geq 20$ J/g criterion provides a separate experimental measure of transformation performance.

Of the 31 compositions that were scaled up and evaluated by UCFTC, the lowest- and highest- ε_{tr} responses are highlighted in Fig. 8. The lowest ε_{tr} was observed for the Iteration 2 alloy Ni₄₆Ti₂₂Co₃Pd₁Hf₂₁Zr₇ (Figs. 8a–c). The raw strain–temperature curves (Fig. 8a) show the expected load dependence of the thermomechanical response, but this alloy also shows an anomalous strain “step” near the upper-cycle temperature (UCT) that was not observed in the other UCFTC-tested materials. Instead of a smooth, approximately linear high-temperature response following transformation, the curve drops abruptly at stresses of

Table 5: Predicted-vs-measured Spearman rank correlation on the 81 transforming alloys synthesized during the campaign (implicit MPE hold-out). Positive ρ indicates rank-ordering preservation; all aggregate (“All”) entries are statistically significant at $p < 10^{-4}$.

Iteration	N	ρ_{M_s}	ρ_{A_f}	$\rho_{\Delta T}$	$\rho_{\Delta H}$
1 (K-medoid, no BO retrain)	23	0.27	0.08	0.09	0.57
2 (BO)	29	0.27	0.55	0.81	0.71
3 (BO)	29	0.73	0.75	0.39	0.66
All	81	0.44	0.56	0.57	0.50

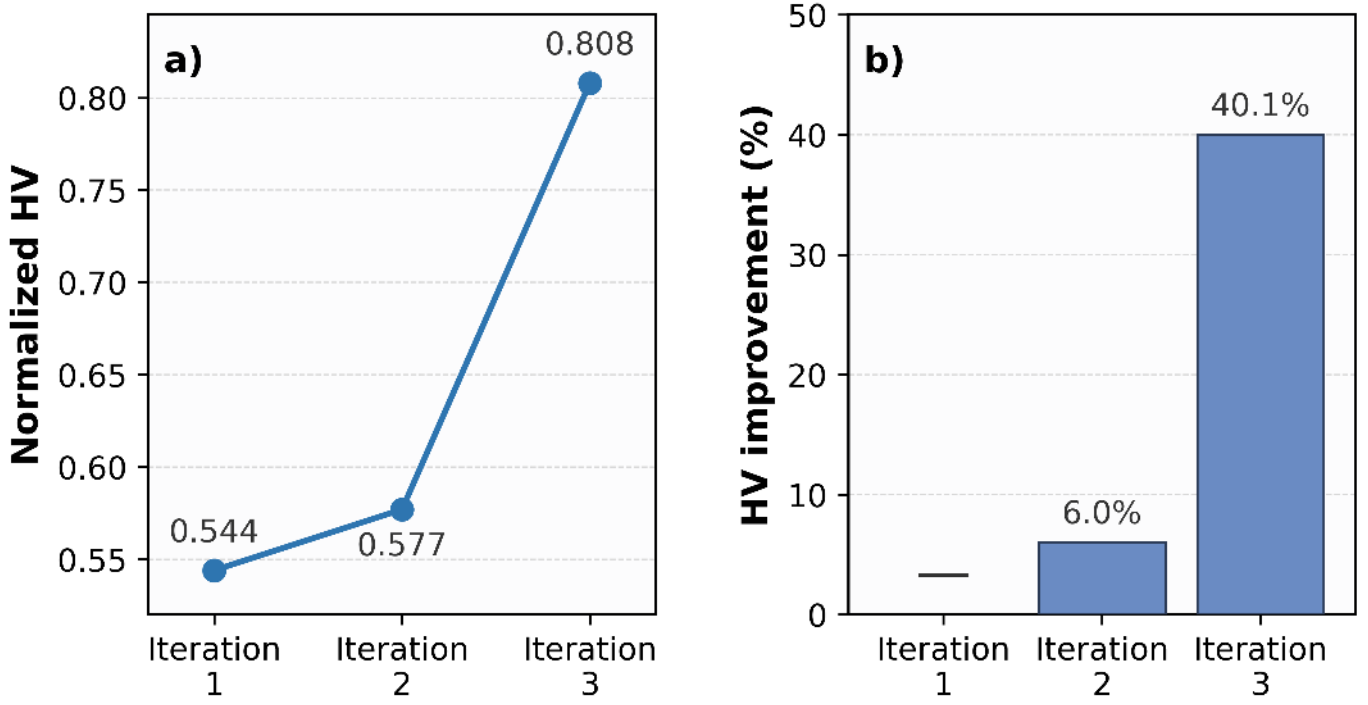


Figure 6: Progress of the BO campaign quantified using the normalized hypervolume (HV) metric for the objective space $(-\Delta T, \Delta H, \epsilon_{tr})$. Hypervolume measures the quality of the current set of non-dominated alloys by simultaneously considering all optimization objectives. (a) Cumulative HV after each optimization iteration, showing progressive expansion of the achievable trade-off space toward lower thermal hysteresis and higher transformation enthalpy and transformation strain. (b) Percentage increase in HV relative to the previous iteration.

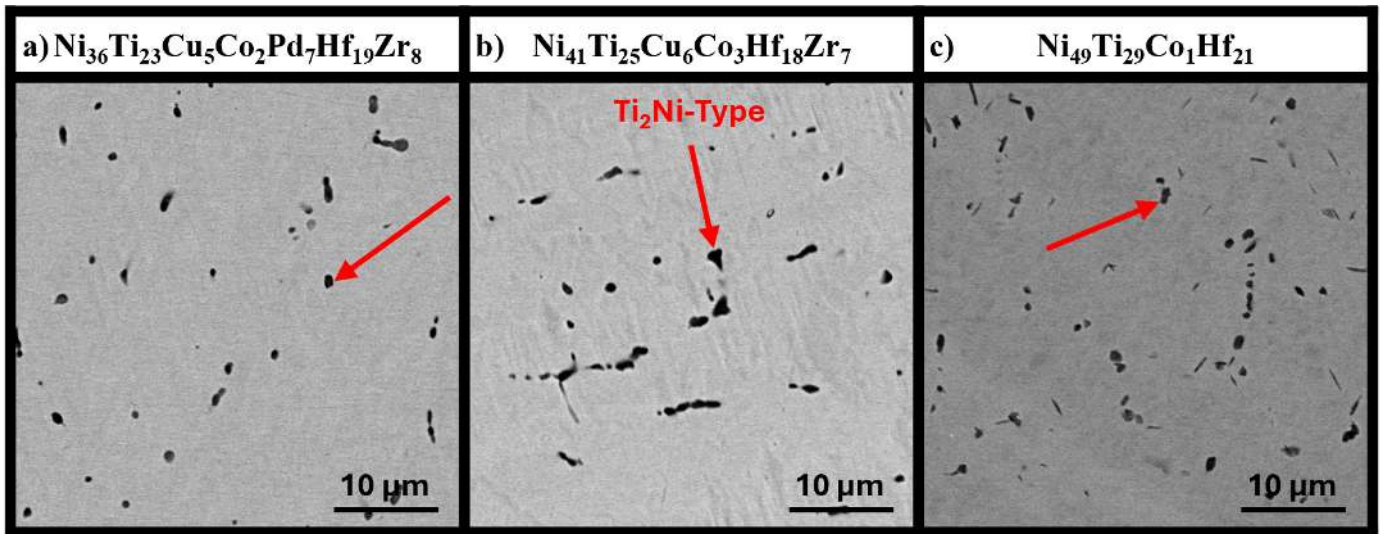


Figure 7: Back-scattered SEM micrographs of selected homogenized alloys. (a–c) Representative microstructures exhibiting a two-phase morphology with B19'-matrix (gray) and dispersed Ti_2Ni secondary phase (black). Ti_2Ni area fractions measured in ImageJ: (a) $\text{Ni}_{36}\text{Ti}_{23}\text{Cu}_5\text{Co}_2\text{Pd}_7\text{Hf}_{19}\text{Zr}_8$, 1.1%; (b) $\text{Ni}_{41}\text{Ti}_{25}\text{Cu}_6\text{Co}_3\text{Hf}_{18}\text{Zr}_7$, 1.3%; (c) $\text{Ni}_{49}\text{Ti}_{29}\text{Co}_1\text{Hf}_{21}$, 1.6%.

150 MPa and above, by as much as 0.17%. The specimen fractured during cooling (below M_f) at 400 MPa. Small perturbations on the cooling traces at all loads are attributed to mechanical vibration from liquid-nitrogen dispensing.

Accordingly, Fig. 8b summarizes poor functional performance: the maximum measured ε_{tr} is only 1.35%, while the residual strain reaches 0.99% at 300 MPa. The ΔT response is likewise unfavorable (Fig. 8c), with a minimum $\Delta T = 66.59^\circ\text{C}$ at 50 MPa (already lower than the stress-free DSC value) and a rapid increase to $\Delta T = 101.83^\circ\text{C}$ by 200 MPa.

In contrast, the best-performing UCFTC alloy was identified in the final iteration: the quinary composition $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ (Figs. 8d–f). Relative to Fig. 8a, the raw curves in Fig. 8d show substantially improved strain recovery and no evidence of the high-temperature step-like strain drop. The enhanced actuation capability is reflected in Fig. 8e, where a maximum $\varepsilon_{tr} = 4.18\%$ is achieved at both 300 MPa and 400 MPa. Increasing the load to 400 MPa, however, increased residual strain from $\sim 0.5\%$ to $\sim 1.2\%$, indicating a trade-off between actuation strain and irrecoverable deformation at the highest applied stress. Finally, Fig. 8f shows that the ΔT at 50 MPa decreases relative to the stress-free DSC value, from 20.4°C to 13.9°C . This improvement is also below the minimum ΔT observed during high-throughput screening (16.5°C), suggesting that modest applied stress can further reduce the effective ΔT for this alloy under UCFTC conditions. The additional ΔT reduction under 50 MPa loading likely combines several effects. Stress-biased variant selection reduces variant-boundary barriers [1, 77]. Mechanical work along the transformation-strain direction lowers the effective activation barrier for both forward and reverse transformations. Distinguishing these contributions would require in-situ diffraction under load. For $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$, the middle-eigenvalue crystallographic compatibility parameter $\lambda_2 = 0.9449$ (computed from the measured B2 and B19' lattice parameters via the framework of [85]), closer to unity than the value for the lowest- ΔT alloy $\text{Ni}_{45}\text{Ti}_{25}\text{Co}_5\text{Hf}_{11}\text{Zr}_{14}$ ($\lambda_2 = 0.9325$). Across the 70 alloys with measured lattice parameters, λ_2 closer to unity is associated with lower stress-free DSC hysteresis (Spearman $\rho = -0.44$, $p < 10^{-3}$), consistent with the λ_2 -hysteresis correlation established in binary and ternary NiTi-based systems. The theoretical single-crystal transformation strain reaches 18.95% along [1,1,1] tension and 12.26% along [1,1,0] tension. The measured polycrystalline 4.18% at 300–400 MPa is well below this envelope, indicating substantial headroom for further optimization via texture control — both computationally, via directional single-crystal calculations, and experimentally, via processing routes that induce preferred orientation. These single-crystal values are not direct predictions of polycrystalline strain, which also depends on grain orientation, grain-to-grain interactions, and the extent of transformation and variant selection, so the comparison alone does not estab-

lish the texture of the specimen. These values and the correlation can be reproduced with the scripts in the public repository [89]. Detailed alloy-specific TEM characterization of $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$ would complement the crystallographic and thermodynamic analysis presented above, but is beyond the scope of this work. Nanoscale precipitation, defect structure, and phase compatibility in closely related Ni-rich NiTi(Hf, Zr) alloys were reported in prior TEM, microstructural, and crystallographic-compatibility studies of this alloy family [28–32, 49, 50, 114, 115].

Direct comparison with the literature is inherently difficult because even small variations in stoichiometry and microstructure can substantially alter alloy behavior. In shape memory alloys, the synthesis route, thermomechanical processing history, and characterization methodology all strongly influence the functional response. A survey of the literature identified 15 distinct alloy systems that have been previously characterized using a range of bulk synthesis and processing routes, as summarized in Fig. 9. Although functional behavior has also been reported in austenitic single crystals and thin films, those responses are not directly transferable to bulk polycrystalline materials and were therefore excluded from the present analysis [12, 116–118].

Figure 9(b) shows that the iterative framework produced MPE HTSMAs with higher transformation strains than any previously reported bulk alloy operating above 250°C . The dense sampling of alloys with martensite start temperatures between 300 and 400°C substantially expanded the known HTSMA design space. Within the literature data, the only alloys reported at transformation temperatures higher than those achieved in this work were the $\text{Ni}_{11}\text{Ti}_{49}\text{Au}_{40}$, $\text{Ni}_{10.3}\text{Ti}_{49.7}\text{Au}_{40}$, $\text{Ni}_{10}\text{Ti}_{50}\text{Au}_{40}$, and $\text{Ni}_9\text{Ti}_{51}\text{Au}_{40}$ alloys studied by Casalena et al. [119]. The functional behavior of those alloys relied on hot extrusion at 900°C with a 7:1 reduction ratio to tailor the mechanical response. In contrast, all alloys developed in the present study are characterized in the as-homogenized condition and define a Pareto front for the simultaneous maximization of transformation temperature and transformation strain. The highest transformation strain obtained in this work is 4.18% at 300 and 400 MPa for $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$.

In addition to high strain, the alloys developed in this study also show narrow thermal hysteresis under load, as shown in Fig. 9(a). $\text{Ni}_{45}\text{Ti}_{25}\text{Co}_5\text{Hf}_{11}\text{Zr}_{14}$ demonstrated a minimum hysteresis of 6.5°C at 50 MPa, which, to the best of the authors' knowledge, is the lowest thermal hysteresis under stress reported for any bulk HTSMA with martensite start temperatures between 150 and 425°C .

Cu was included in the design space (Section 2.3) because of its documented hysteresis reduction at $\geq 15\text{at.}\%$ via a B19' $\rightarrow$ B19 martensite transition [15, 53, 54]. The two lowest-hysteresis alloys identified here — $\text{Ni}_{45}\text{Ti}_{25}\text{Co}_5\text{Hf}_{11}\text{Zr}_{14}$ and $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$, both Cu-free — instead achieve low ΔT through Hf/Zr and small Pd substitutions without requiring high-Cu compositions.

However, the present results do not establish the separate contributions of the constituent elements or demonstrate that improved crystallographic compatibility alone accounts for the observed hysteresis.

The functional properties achieved in the NiTi(Co,Cu,Pd,Hf,Zr) alloys reported here arise solely from compositional tuning, without the use of thermo-mechanical processing. Of the 899 literature data points shown in Fig. 9, only 21 (2.3%) correspond to alloys characterized in the as-homogenized condition, and in this case, they are all NiTiHf and NiTiPd. In comparison, 31 alloys were developed and characterized in the as-homogenized state in the present study, representing the largest expansion to date of homogenized mechanical characterization in bulk HTSMAs. Incorporating thermo-mechanical processing into the MPE HTSMAs identified here therefore remains an important direction for future optimization of alloy performance.

4. Summary and Conclusions

This work developed and experimentally demonstrated a closed-loop, multi-objective batch Bayesian optimization framework for the discovery of NiTi(Co,Cu,Pd,Hf,Zr) MPE HTSMAs. Starting with a literature-informed design space containing 3,812,408 candidate compositions, CALPHAD-based phase-stability screening reduced the search space to 17,207 feasible alloys. Three optimization iterations comprising 87 synthesized alloys and 31 thermo-mechanically characterized scaled-up alloys were completed. Thus, only 0.0023% of the original 3,812,408-composition design space was experimentally synthesized, yet the campaign still identified high-performing alloys and expanded the Pareto-optimal property space. Across the campaign, the normalized hypervolume increased from 0.544 to 0.808, corresponding to a 48.5% expansion of the accessible multi-objective design space and demonstrating the effectiveness of the closed-loop optimization framework.

The main findings of this study can be summarized as follows:

- The BBO framework progressively concentrated sampling within martensitically transforming composition regions of the design space. All six non-transforming alloys identified during the campaign were discovered during the exploration-oriented first iteration, whereas all alloys selected in Iterations 2 and 3 showed martensitic transformation. Under a CALPHAD-only baseline calibrated on Iteration 1, the probability of observing zero non-transforming alloys in 58 independent CALPHAD-feasible draws is $\sim 10^{-6}$, indicating that the ML + BO loop concentrates sampling within the transformation-capable subregions of the CALPHAD-feasible space rather than discovering them by chance.

- The iterative optimization campaign systematically improved alloy performance. The median thermal hysteresis decreased from 45.0 °C in Iteration 1 to 32.0 °C in Iteration 3, while the median transformation enthalpy increased from 22.0 to 26.6 J/g. Simultaneously, the fraction of alloys satisfying the target M_s window (200–400 °C) increased from 41% to 72%.
- Increasing elemental complexity was associated with a consistent reduction in transformation enthalpy. Transformation temperatures did not show a comparable monotonic trend and were reduced only at the septenary level. Septenary alloys showed the lowest median M_s and ΔH values and did not produce any fully qualified compositions. Instead, the strongest property balance emerged in lower-complexity alloys, particularly quinary compositions, indicating that “more elements” is not inherently better for MPE HTSMAs.
- Composition–property analysis identified Ti-site substitution by Hf and Zr as the primary mechanism for achieving elevated transformation temperatures, while cumulative Ni-site substitution (Cu + Co + Pd) generally reduced both transformation temperature and transformation enthalpy under the fixed processing condition used in this study.
- Among the 31 scaled-up alloys evaluated by uniaxial constant-force thermal cycling, the quinary alloy Ni₄₆Ti₂₈Co₂Pd₂Hf₂₂ emerged as the highest-performing composition. This alloy showed the lowest thermal hysteresis among all fully qualified alloys ($\Delta T = 16.5$ °C), the highest measured recoverable transformation strain ($\varepsilon_{tr} = 4.18\%$), and a transformation enthalpy of approximately 28 J/g while maintaining a martensite start temperature near 225 °C. These results demonstrate that low-hysteresis, high-strain actuation can be achieved in this MPE alloy family using only 2 at.% Pd. This result also identifies modest Pd additions of approximately 2–3 at.% as an effective strategy for reducing hysteresis without relying on the high Pd contents commonly used in conventional NiTiPd HTSMAs. Comparison against 899 literature data points (Fig. 9) confirms that Ni₄₆Ti₂₈Co₂Pd₂Hf₂₂ outperforms every previously reported bulk HTSMA in the $200 \leq M_s \leq 400$ °C window on the combined (ΔT , ε_{tr}) objectives.
- Quinary Ni–Ti–Co–Pd–Hf alloys gave the best joint balance of M_s , ΔT , ΔH , and ε_{tr} in this campaign, with Ni₄₆Ti₂₈Co₂Pd₂Hf₂₂ the specific composition satisfying all four targets simultaneously.

Overall, this work demonstrates that closed-loop, metallurgy-informed BO can substantially accelerate HTSMA discovery. The campaign returned 20 fully qualified alloys, whereas uninformed random sampling of 87

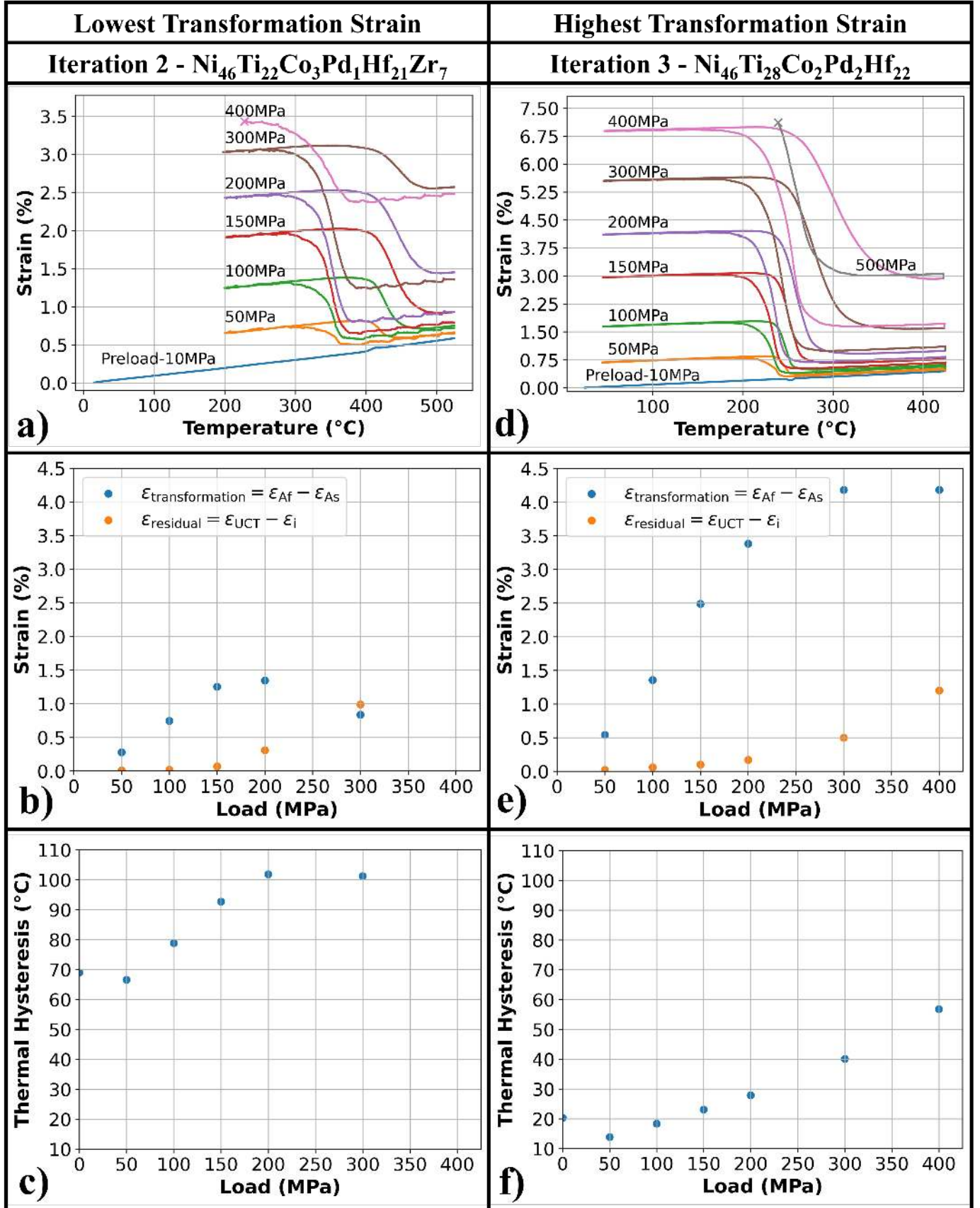


Figure 8: UCFTC comparison of the lowest- and highest-transformation-strain alloys among the 31 scaled-up compositions. (a–c) Iteration 2 $\text{Ni}_{46}\text{Ti}_{22}\text{Co}_3\text{Pd}_1\text{Hf}_{21}\text{Zr}_7$, exhibiting limited recoverable ϵ_{tr} , high residual strain accumulation, and fracture during cycling at 400 MPa. (d–f) Iteration 3 alloy $\text{Ni}_{46}\text{Ti}_{28}\text{Co}_2\text{Pd}_2\text{Hf}_{22}$, exhibiting stable thermomechanical response, improved strain recovery, ϵ_{tr} up to 4.18% at 300–400 MPa, and reduced ΔT under 50 MPa relative to the stress-free DSC response. Small perturbations in the cooling segments of the strain–temperature curves arise from mechanical vibrations associated with the liquid-nitrogen dispensing valve.

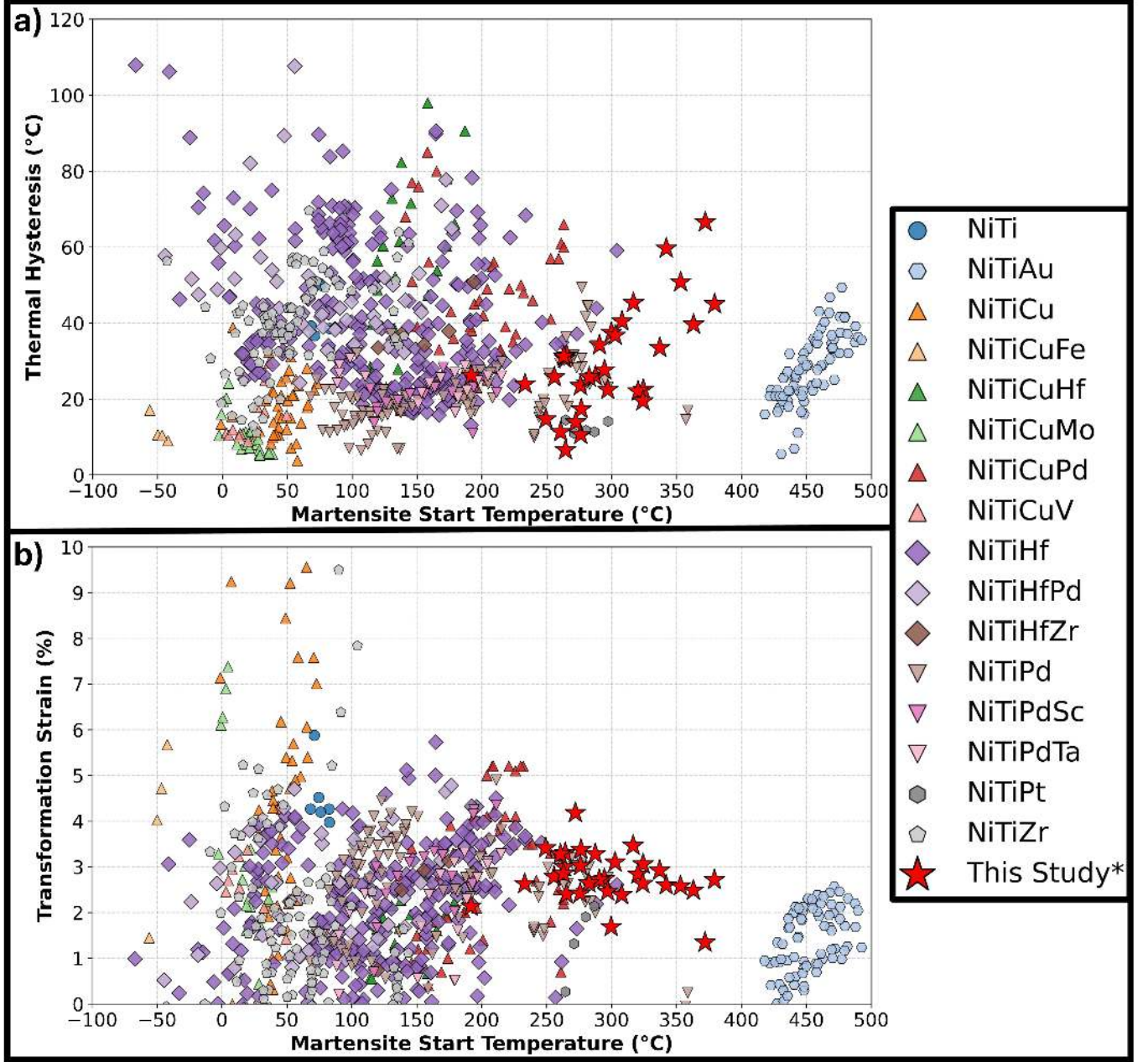


Figure 9: Constrained literature comparison of martensite start temperature (M_s) and key performance metrics for bulk HTSMAs, including the scaled-up alloys developed in this work. To ensure a meaningful comparison, the literature dataset was restricted to alloys produced using bulk metallurgical processing routes (e.g., vacuum arc melting, vacuum induction melting, and Ar-arc melting), while single-crystal and thin-film studies were excluded. (a) Minimum thermal hysteresis (ΔT) measured under load as a function of M_s ; (b) maximum transformation strain (ϵ_{tr}) measured under load as a function of M_s . The comprehensive citation list acknowledges every research group whose published data made this comparison possible. References for the literature dataset are provided in [1, 12, 24, 28–31, 36, 41, 49, 53, 54, 77, 78, 93, 114, 115, 119–325].

unconstrained compositions from the 3.8-million-alloy design space would be expected to return fewer than one — a joint $\sim 230\times$ efficiency gain arising from CALPHAD phase-stability screening in series with ML + BO acquisition (see Section 3 for the attribution analysis and the argument for treating the two components as synergistic rather than independent). Meaningful improvements in HTSMA performance can be achieved through compositional optimization alone under a fixed processing schedule — here, homogenization at 950 °C for 24 h followed by water quench — but the framework does not, in its present form, discriminate among alloys whose performance depends more strongly on precipitation heat treatment or thermomechanical processing. Because different compositions can develop different microstructures under the same treatment, the reported performance remains conditional on this processing history. Extending the acquisition to include processing variables is feasible and has been demonstrated for the lower-dimensional NiTiCuHf system [82], where composition and processing were jointly optimized. In the present multi-principal-element setting, processing was fixed to keep the composition search tractable. Future efforts should couple processing variables and higher-resolution microstructural characterization directly with the closed-loop learning process, enabling the transition from composition-only discovery toward integrated composition-processing optimization of robust, application-ready HTSMAs.

Supplementary Materials

The following supplementary files accompany this manuscript:

- **Supplementary Information S1** (Supplemental_Information_Experimental_Data.xlsx) – tabulated compositions, processing conditions, DSC-derived transformation temperatures and enthalpies, XRD lattice parameters, and ML-predicted properties for all alloys generated across the three Bayesian-optimization iterations.
- **Supplementary Information S2** (Supplemental_Information_UCFTC_Curves.pdf) – raw uniaxial constant-force thermal cycling (UCFTC) strain-temperature curves and tangent-intercept analyses for the 31 scaled-up compositions.

Data and Code Availability

Code and data for the three property predictors used in this work are publicly available on GitHub: transformation-temperature CatBoost model (<https://github.com/sinazadeh/CatBoost-SMAs>); NiTi phase-compatibility λ_2 model (<https://github.com/sinazadeh/Phase-Compatibility-Model-NiTi>); and the LDT-based transformation-strain model (<https://github.com/sinazadeh/Transformation-Strain-Model-NiTi>).

The notebook and helper code used to select features for and train the ΔH CatBoost model are available at <https://github.com/sinazadeh/NiTi-alloy-discovery> (ML models directory). The full composition, processing, DSC, XRD, and ML-prediction record for the 87 synthesized and 31 scaled-up alloys is provided as Supplementary Information S1. A complete reproducibility bundle for this campaign is archived on Zenodo [89] and mirrored at https://github.com/rarroyave/2026_Zadeh-HTSMA_public. The bundle contents and their relationship to specific manuscript results are documented in Appendix B.

Declaration of competing interest

The authors have no financial or personal connections that could have appeared to influence the work reported in this paper.

Declaration of generative AI in scientific writing

Generative AI tools (ChatGPT 5.2 and Gemini 3) were used to assist with drafting and language editing of selected portions of this manuscript (e.g., improving clarity, grammar, and flow). The authors reviewed, verified, and edited all AI-assisted text and accept full responsibility for the content, interpretations, and conclusions presented.

CRedit authorship contribution statement

Sina Hossein Zadeh: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Validation, Visualization, Writing – original draft. **John Broucek:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Validation, Visualization, Writing – original draft. **Cem Cakirhan:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Validation, Visualization, Writing – original draft. **Mingqian Li:** Methodology, Writing – original draft. **Danial Khatamsaz:** Methodology, Software, Investigation. **Xiaoning Qian:** Conceptualization, Writing – review & editing, Supervision. **Ibrahim Karaman:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Raymundo Arroyave:** Conceptualization, Writing – review & editing, Supervision, Project administration.

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Appendix A: Quantifying the methodological advantage

A.1 Setup

This appendix quantifies how often a campaign as successful as the present one would arise if alloys were instead drawn at random from the unconstrained MPE design space of $N_{\text{tot}} = 3,812,408$ candidates — the null model H_0 . Call a composition *fully qualified* if it simultaneously satisfies six conditions: (i) phase-pure single-B2 austenite at the homogenization temperature ($> 97\%$); (ii) measurable martensitic transformation under the standard thermal protocol; (iii) $200 \leq M_s \leq 400^\circ\text{C}$; (iv) $\Delta T \leq 50^\circ\text{C}$; (v) $\Delta H \geq 20 \text{ J/g}$; and (vi) $\varepsilon_{\text{tr}} \geq 2.5\%$. Let p_q denote the chance that a randomly chosen composition meets all six. The hit rates used to calibrate these estimates can be reproduced with the scripts in the public repository [89].

A.2 Estimating the baseline probability

The probability p_q is split into two factors,

$$p_q = p_C \cdot p_{4|C}, \quad (\text{A.1})$$

where p_C is the chance a random composition passes CALPHAD phase-purity screening and $p_{4|C}$ is the chance that — given it did pass — it also satisfies all four property targets (M_s window, ΔT , ΔH , ε_{tr}). The first factor is exact from the CALPHAD filter:

$$p_C = \frac{17,207}{3,812,408} \approx 4.5 \times 10^{-3}. \quad (\text{A.2})$$

The second factor is estimated from Iteration 1, the only campaign batch built without BO guidance (K-medoid clustering drew a maximally diverse sub-batch of the 17,207 CALPHAD-feasible candidates). Of those 29 alloys, 6 satisfied all four targets, so

$$\hat{p}_{4|C} = \frac{6}{29} \approx 0.21, \quad (\text{A.3})$$

and

$$\hat{p}_q \approx 4.5 \times 10^{-3} \cdot 0.21 \approx 1 \times 10^{-3}. \quad (\text{A.4})$$

In plain terms, on average only about 1 of every 1,000 random compositions in this MPE design space would simultaneously meet all six conditions.

A.3 Tail probability under unconstrained random sampling

Let X be the number of fully qualified alloys produced by $n = 87$ independent random draws under H_0 . Then $X \sim \text{Binomial}(n, p_q)$, and because the expected number $\lambda = np_q = 0.087$ is much smaller than one, the Poisson approximation $X \sim \text{Poisson}(\lambda)$ is accurate. The campaign delivered 20 confirmed all-four-pass alloys (Table 3), so the chance of matching or exceeding this yield purely by random sampling is

$$\Pr(X \geq 20 | H_0) = \sum_{k=20}^n \binom{n}{k} p_q^k (1 - p_q)^{n-k} \approx \frac{e^{-\lambda} \lambda^{20}}{20!} \approx 2 \times 10^{-40}. \quad (\text{A.5})$$

Equivalently, the expected random-sampling yield is $E[X] = 0.087$ alloys, so the observed 20 represents a $\sim 230\times$ gain. This tail probability should be interpreted as an order-of-magnitude sanity check rather than a rigorous p-value. The conditional success rate, $p_{4|C} = 6/29$, is estimated from a single 29-alloy batch selected by K-medoid clustering rather than by uniform random sampling. The non-transforming rate q used in §A.5 is estimated from that same batch but counts a different set of alloys — 6/29 non-transforming, not the 6/29 that pass all four targets. The analyses in §A.3 and §A.5 are therefore complementary consistency checks on one calibration dataset, not independent statistical confirmations.

A.4 Robustness to the conditional-rate estimate

The estimate $\hat{p}_{4|C} = 6/29$ comes from only 29 alloys, so it carries its own uncertainty. The 95% Wilson confidence interval for 6/29 is $[0.10, 0.38]$; recomputing the tail probability at the bounds of this interval (Table A1) gives probabilities ranging from $\sim 10^{-47}$ to $\sim 10^{-35}$. In short, even under the most generous assumption (a 38% conditional success rate), random sampling still cannot plausibly produce the observed yield.

Table A1: Tail probabilities $\Pr(X \geq 20 | H_0)$ for the unconstrained random-sampling null model at point estimate and 95% Wilson CI bounds of the conditional rate $\hat{p}_{4|C}$.

	Lower CI	Point estimate	Upper CI
$p_{4 C}$	0.10	0.21	0.38
p_q	4.5×10^{-4}	9.5×10^{-4}	1.7×10^{-3}
$\Pr(X \geq 20 H_0)$	$\sim 10^{-47}$	$\sim 2 \times 10^{-40}$	$\sim 10^{-35}$

A.5 The CALPHAD-only comparison

The unconstrained-random baseline of §A.2 includes the $\sim 220\times$ reduction from the CALPHAD filter, which cannot be cleanly separated from the ML + BO contribution (Section 3). A more demanding test of the closed-loop ML + BO contribution restricts the baseline to random sampling *from the CALPHAD-feasible subset* and focuses on the property measured on every synthesized alloy: transformability. Let q denote the non-transforming rate in the CALPHAD-feasible subset, estimated from the same K-medoid Iteration 1 batch as $p_{4|C}$ (§A.2), $\hat{q} = 6/29 \approx 0.21$ (95% CI $[0.10, 0.38]$). For $m = 58$ such draws (the combined size of Iterations 2 and 3), the probability of seeing zero non-transformers is

$$\Pr(Y = 0 | H'_0) = (1 - q)^m. \quad (\text{A.6})$$

The campaign observed $Y = 0$. Table A2 reports this probability at the point estimate and CI bounds: at the

point estimate, the chance is $\sim 10^{-6}$ — roughly one in a million — and even at the most generous lower CI bound it remains below 0.3%.

Table A2: Probability $\Pr(Y = 0 \mid H'_0)$ that 58 uniform random CALPHAD-feasible draws yield zero non-transforming alloys, at the point estimate and 95% Wilson CI bounds of the non-transforming rate $\hat{q}$.

	Lower CI	Point estimate	Upper CI
q	0.10	0.21	0.38
$\Pr(Y = 0 \mid H'_0)$	2.2×10^{-3}	1.2×10^{-6}	9.4×10^{-13}

A.6 Pareto-front contribution by iteration

The all-four-pass count (§A.2–A.4) and the transformability count (§A.5) treat the design targets as binary pass/fail criteria. A complementary measure of BO progress is the per-iteration contribution to the cumulative Pareto front in the $(-\Delta T, \Delta H, \varepsilon_{\text{tr}})$ objective space (here $\Delta T = A_{50} - M_{50}$ from the second DSC cycle, ΔH the second-cycle average enthalpy, and ε_{tr} the largest measured transformation strain), which captures multi-objective trade-offs explicitly. Of the 31 strain-tested alloys (10 from Iteration 1, 11 from Iteration 2, 10 from Iteration 3), 10 lie on the final cumulative Pareto front. Table A3 reports the per-iteration breakdown.

The Iteration 3 per-alloy rate (0.50) is $2.5\times$ the Iteration 1 rate (0.20). If BO offered no advantage over random K-medoid seeding, the number Z of Pareto-optimal alloys in $m = 10$ Iteration 3 draws would follow $Z \sim \text{Binomial}(10, 0.20)$, and

$$\Pr(Z \geq 5 \mid H''_0) = \sum_{k=5}^{10} \binom{10}{k} (0.20)^k (0.80)^{10-k} \approx 0.033. \quad (\text{A.7})$$

In plain terms, even under the null hypothesis that BO offers no advantage, the observed Iteration 3 Pareto contribution would arise by chance only about once in 30 trials.

Iteration 3 also refined the prior front: after Iteration 2 the cumulative front contained 8 alloys; Iteration 3 added 5 new Pareto-optimal alloys and strictly dominated 3 of the earlier ones, removing them from the front. The closed-loop BO is therefore not just expanding the Pareto frontier but actively improving it.

A.7 Caveats

Three limitations qualify these estimates. (i) Iteration 1 used K-medoid diverse seeding, not strictly uniform random sampling. K-medoid spreads its 29 draws more evenly across the descriptor space, so $\hat{p}_{4|C}$ and $\hat{q}$ may over- or underestimate the true random-in-feasible rates; the tail probabilities in §A.3 and §A.5 are therefore order-of-magnitude bounds anchored on this calibration, not rigorous p-values. (ii) The CALPHAD pass rate $p_C = 0.45\%$ is specific to this design space and the TCNI12 database and

would shift modestly under alternative thermodynamic assessments. (iii) The ML surrogates were trained primarily on binary–quaternary NiTi data. On the MPE alloys synthesized in this campaign, their pointwise errors are larger than on the original datasets, as expected for compositions outside the training domain, while their rank correlations with the measured values are positive and significant for all four properties and increase once the models are re-trained on campaign data (Table 5). The surrogates are therefore used to rank candidates for the acquisition function rather than as quantitative predictors of MPE properties. Against λ_2 values computed from the measured lattice parameters of 70 alloys, the λ_2 model gives a rank correlation of 0.49 ($p < 10^{-4}$) and a mean absolute error of 0.012, close to its original accuracy, and the LDT strain model reproduces the dependence of theoretical strain on loading direction. These analyses are accordingly order-of-magnitude statements that defend the qualitative claim of a large methodological advantage, not precise estimators of campaign performance.

Appendix B: Reproducibility Bundle and Machine-Learning Surrogate Details

The data and code needed to reproduce the results in Section 3 and Appendix A are archived on Zenodo [89] (DOI: 10.5281/zenodo.22256930) and on GitHub (https://github.com/rarroyave/2026_Zadeh-HTSMA_public). The repository installs in a standard Python environment and regenerates the reported results; automated tests on Python 3.10, 3.11, and 3.12 confirm this after each update.

The repository contains the dataset for all 87 alloys (compositions, homogenization schedule, DSC transformation temperatures, XRD lattice parameters, UCFTC data per ASTM E3097, and ML predictions), identical to Supplementary Information S1, and five Python scripts reproducing the main numerical results: the per-iteration Spearman correlations and mean-absolute errors of Table 5; the per-iteration functional hit-rates anchoring the Appendix A calibration; the champion-alloy Ball–James parameters $(\lambda_1, \lambda_2, \lambda_3)$ and single-crystal transformation strains along the five directions of Section 2.5; the comparison of the λ_2 and LDT strain models with values computed from the measured lattice parameters of 70 alloys, together with the measured- λ_2 /DSC-hysteresis correlation (Section 3); and the per-iteration Pareto-front membership and its tail probability (Section 3 and Appendix A.6). A Jupyter notebook, provided with its outputs, reproduces the first three and plots the iteration progression, the hit-rate trends, and the per-direction compatibility parameters.

The three publicly released predictors — the CatBoost transformation-temperature model [84], the composition-based λ_2 phase-compatibility model [85], and the LDT-based transformation-strain model [86] — are included as archived copies of specific versions, so the repository remains usable if the originals change. The single-output

Table A3: Per-iteration Pareto-front membership in the $(-\Delta T, \Delta H, \varepsilon_{tr})$ objective space, computed over the 31 strain-tested alloys with all three properties measured.

Iteration	Strain-tested	On final Pareto front	Per-alloy rate
1 (K-medoid seeding, no BO)	10	2	0.20
2 (BO-guided)	11	3	0.27
3 (BO-guided)	10	5	0.50
Total	31	10	0.32

CatBoost model for ΔH (Section 2.5) is not released as a trained model, but the notebook and helper code used to select its features and train it accompany the Bayesian optimization code; its training data are not included, and its predictions for the synthesized alloys are tabulated in Supplementary Information S1. The Bayesian optimization code is provided with the data from each iteration (feasibility labels, tested alloys, and measured objective values) and the Gaussian-process models, multi-objective EHVI acquisition, and reification-based information fusion used to select each batch. A short script regenerates the full 3,812,408-composition candidate space (Section 2.3) in about 15s, identical to the campaign list, so the 128 MB file is not included.

All three released surrogates use CBFV composition-based featurization with the `jarvis` elemental-property set. Beyond CBFV, the transformation-temperature predictor adds alloy-family categorical flags (`niti_bas`, `nitihf_bas`, `nitipd_bas`, `niticu_bas`, `nitinb_bas`, `nitizr_bas`, `nitib_bas`) and processing descriptors (homogenization temperature and time); the λ_2 model adds HEACalculator-derived geometric-strain descriptors; and the LDT + CatBoost strain model treats deformation type as a categorical feature and deformation direction as an embedding feature. The ΔH model uses six descriptors chosen by CatBoost’s built-in feature selection from CBFV features (`jarvis`, `magpie`, `oliynyk`, `mat2vec`, and one-hot element sets), mixing descriptors computed with HEACalculator (<https://github.com/dogusariturk/HEACalculator>), and element-ratio combinations, after removing one feature from each pair with an absolute correlation above 0.9.

Table B1 summarizes the key training hyperparameters and cross-validation protocols. After each campaign iteration, the transformation-temperature and ΔH CatBoost predictors were retrained with the newly measured MPE data appended. The λ_2 and LDT models were used as fixed deterministic filters throughout the campaign.

As a supplementary check, the repository also trains ΔH models on the measured campaign alloys alone. In cross-validation these reach a mean absolute error of about 3–4 J/g, but they do not rank the alloys of the next iteration better than the campaign model, whose accuracy on the campaign alloys is reported in Table 5.

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Table B1: Key hyperparameters and cross-validation protocols for the four ML surrogates used in this work. See Data and Code Availability for full implementations.

Model	Type	Loss	LR	Depth	L_2 reg	CV protocol
Transformation temperatures λ_2 phase compatibility	CatBoost (multi-output) Composition-based analytical + regression	MultiRMSE —	0.209 —	6 —	0.060 —	80/20 <code>use_best_model</code> $R^2 = 0.91$ on hold-out [85]
Theoretical ε_{tr} (LDT)	CatBoost (single-output)	MAE	default	default	default	10-fold <code>random_state=42</code> KFold,
Transformation enthalpy ΔH	CatBoost (single-output)	MAE	default	default	default	Not recorded; accuracy on campaign alloys in Table 5

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